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Thermodynamics of rare earth sesquioxides

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Abstract

Experimental thermochemical and related data for rare earth sesquioxides, R_2O_3 (R = Sc, Y, La–Lu) are critically reviewed to provide a consistent set of thermodynamic functions covering the temperature range from at least 298 K up to above the melting point taking the polymorphism into account. Discrepancies in the available data are discussed and further studies to resolve them and to fill the gaps in the current knowledge are suggested. The best values for the temperatures of the reversible polymorphic transformations are recommended. Using high-pressure and molar volume data as well as ionic radius of a trivalent rare earth cation as an independent parameter the relative stabilities of different structures (the cubic C-form, the monoclinic B-form, the hexagonal A-form, the hexagonal H-form, and the cubic X-form) are estimated quantitatively for all rare earth sesquioxides. Based on the empirical relationship between the entropy of fusion and the corresponding fractional volume change, the entropies and enthalpies of fusion of the lanthanide sesquioxides are estimated. The reliability of the assessed functions and transition data is tested through the sample calculations of R'_2O_3–R''_2O_3 phase diagrams, where R′ and R″ represent different rare earth cations.
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1. Introduction

Rare earth sesquioxides (R_2O_3, R = lanthanide, Y, or Sc) possess unique physical and chemical properties, which make them technologically very important and many review articles dealing with fundamentals and applications of these materials have already been published. The crystallographic data and polymorphism of the rare earth sesquioxides

were first reviewed by Brauer [1] and updated by Haire and Eyring [2]. Depending on the temperature and the radius of R^{+3}-cation rare earth sesquioxides are known to occur in five polymorphic modifications, A-, B-, C-, H-, and X-R_2O_3. Their crystal structures are visualized in Figs. 1.1–1.5. In the C-type structure all cations are in six-fold coordination (Fig. 1.1), in the B-type structure some of them are surrounded by seven oxygen anions (Fig. 1.2). This seven-fold coordination is a characteristic feature of the A-type structure (Fig. 1.3). Both the H- and X-type structures are built from octahedra (Figs. 1.4 and 1.5), which are distorted in the former case.

At normal conditions, hexagonal A-form is the stable structure for R = La, Ce, Pr, Nd, while Sm_2O_3, Eu_2O_3, and Gd_2O_3 may exist in both monoclinic B- and cubic C-form. The C-type structure is characteristic for R = Tb, Dy, Ho, Er, Tm, Yb, Lu, Y, and Sc. The A-, B-, or C-R_2O_3, except Lu_2O_3 and Sc_2O_3 transform into hexagonal H-phase at sufficiently

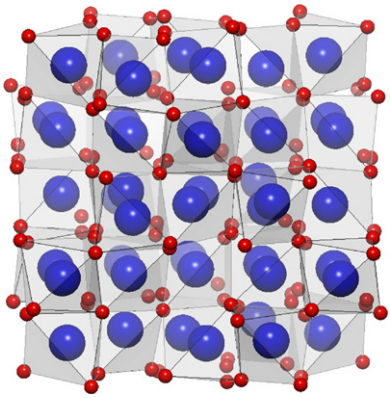

Fig. 1.1. The polyhedral representation of the cubic C-type structure of R_2O_3. The large and small spheres stand for R and O atoms, respectively.

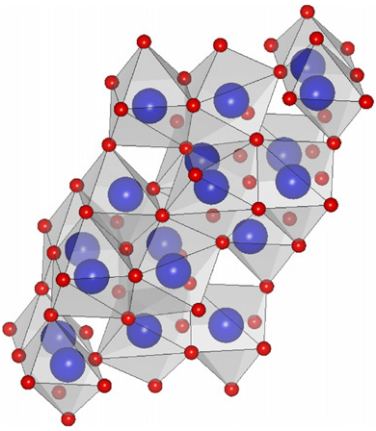

Fig. 1.2. The polyhedral representation of the monoclinic B-type structure of R_2O_3. The large and small spheres stand for R and O atoms, respectively.

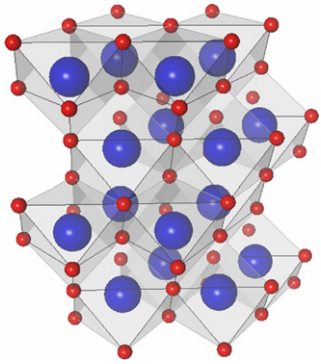

Fig. 1.3. The polyhedral representation of the hexagonal A-type structure of R_2O_3. The large and small spheres stand for R and O atoms, respectively.

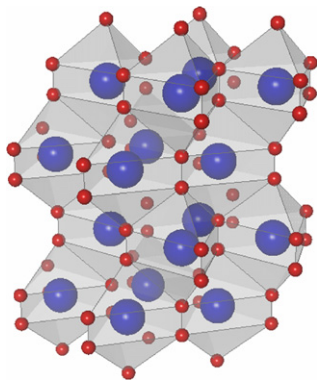

Fig. 1.4. The polyhedral representation of the high-temperature hexagonal H-type structure of R_2O_3. The large and small spheres stand for R and O atoms, respectively.

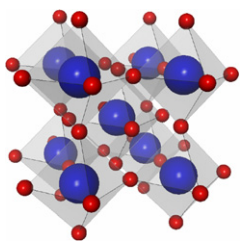

Fig. 1.5. The polyhedral representation of the high-temperature cubic X-type structure of R_2O_3. The large and small spheres stand for R and O atoms, respectively.

high temperatures. The latter converts into high-temperature cubic X-R_2O_3 for R = La–Dy before the melting occurs. Foex and Traverse [3] suggested that the C-form of La_2O_3, Ce_2O_3, Pr_2O_3, and Nd_2O_3 is actually stable at room temperature and transforms into A-form upon heating. Recent study by Fedorov et al. [4] has shown, however, that

the C ⇒ A transition in La_2O_3 and Nd_2O_3 is accompanied by the removal of volatile impurities and thus, is not a polymorphic transition. The thermodynamic stability of C-type structure for Sm_2O_3, Eu_2O_3, and Gd_2O_3 at low temperatures is still under debate.

Knowledge of thermodynamic properties of rare earth sesquioxides is of great significance. Holley at al. [5] reviewed the literature on the enthalpies, entropies and heat capacities of rare earth oxides before 1966 and tabulated the thermodynamic functions for A-type R_2O_3 (R = La–Nd), B-type R_2O_3 (R = Sm–Gd), and C-type R_2O_3 (R = Sm–Lu, Y, Sc). An up-to-date review of the enthalpies of formation of lanthanide sesquioxides has recently been provided by Cordfunke and Konings [6]. In their work, a critical assessment of the published data has been made including a careful analysis of measurements conditions, purity of the samples and auxiliary data used in thermochemical cycle for calculations. Gruber et al. [7] analyzed the data on heat capacity and heat content of the A-type La_2O_3, Ce_2O_3, Pr_2O_3, and Nd_2O_3 and obtained a consistently assessed curve (functional basis $T = 0$ K) covering the entire temperature range reported for those compounds. Kriklya [8] estimated the heat capacities of A-R_2O_3 (R = Pm, Sm, Eu), B-R_2O_3 (R = Ce–Pm, Tb–Ho), and C-R_2O_3 (R = Ce–Pm) on the basis of the experimental data on the heat capacity of A-La_2O_3, B-Gd_2O_3, and C-Gd_2O_3 and calculated Schottky contributions. Most of those phases are metastable and no corresponding measurements are reported.

The purpose of the present review is to provide a consistent set of thermodynamic functions for rare earth sesquioxides covering the temperature range from at least 298 K up to above the melting point taking the polymorphism into account. For the metastable modifications, the energetic differences with respect to stable structures are estimated.

2. Assessed temperatures of phase transitions

2.1. Transitions at normal pressure

The reported temperatures of the reversible solid ⟺ solid phase transitions in rare earth sesquioxides at ambient pressure ($P = 1$ bar) are compiled in Table 2.1. Sc_2O_3 and Lu_2O_3 exist only in the cubic form up to the melting point, while the available values for Pm_2O_3 are very uncertain [9]. Also, the temperatures of C ⟺ B phase transitions in Sm_2O_3, Eu_2O_3, and Gd_2O_3 reported in numerous studies are not included in Table 2.1. Apart from the huge scatter all they suffer from the slow kinetics. Some of these data are shown in Fig. 2.1. The melting points of rare earth sesquioxides were already reviewed [10,11] and the results of these critical assessment works are accepted here without modifications. The transformation scheme for lanthanides sesquioxides is presented graphically in Fig. 2.1. Systematic studies across the whole lanthanide series have been carried out by two groups. The French group [3,12] used thermal analysis coupled with a solar furnace in air. Thermal arrests corresponding to the polymorphic transformation were registered on cooling. The formation of a particular phase was confirmed by *in situ* high-temperature X-ray diffraction in He or He + H_2 atmosphere. The Ukrainian group [13,14] utilized the differential thermal analysis (DTA) in tungsten crucibles under protective atmosphere (Ar/He), while thermal effects were read from heating curves.

The selected values (Table 2.2) are based on the results of these four works and are divided into three categories: recommended values (*R*), where all measured temperatures are reasonably consistent and precise, probable values (*P*), where the consensus of experimental values is less convincing, and estimated values (*E*), where experimental data are

Table 2.1
Reported temperatures of the reversible solid ⟺ solid phase transitions (K) in rare earth sesquioxides at ambient pressure ($P = 1$ bar)

Oxide	Transition						Ref.
	C ⟺ B	B ⟺ A	A ⟺ H	B ⟺ H	C ⟺ H	H ⟺ X	
Y_2O_3					2583		[3]
					2573		[12]
					2623 ± 20		[13]
					2663 ± 30		[14]
					2600 ± 30		[16]
					2565		[17]
					2573		[18]
					2623 ± 20		[19]
					2512 ± 25		[20][a]
					2581 ± 15		[21][a]
La_2O_3			2313			2383	[3]
			2313			2373	[12]
			2323 ± 20			2413 ± 20	[13]
			2303 ± 30			2373 ± 30	[14]
			2313			2363	[22]
Ce_2O_3			2393 ± 20			2443 ± 20	[13]
			2393 ± 30			2443 ± 30	[14]
Pr_2O_3			2223			2423	[3]
			2318 ± 20			2393 ± 20	[13]
			2303 ± 30			2403 ± 30	[14]
Nd_2O_3			2373			2473	[3]
			2373			2473	[12]
			2373 ± 20			2473 ± 20	[13]
			2333 ± 30			2453 ± 30	[14]
			2373 ± 10			2473 ± 10	[23]
			2340			2435	[17]
Sm_2O_3		2173	2373			2523	[3]
		2073				2523	[12]
		2193 ± 20	2403 ± 20			2553 ± 20	[13]
		2143 ± 30	2343 ± 30			2498 ± 30	[14]
		2203	2433				[24]
Eu_2O_3		2323	2413			2543	[3]
		2323 ± 20	2413 ± 20			2543 ± 20	[19]
Gd_2O_3		2443	2473			2633	[3]
		2353				2633	[12]
		2443 ± 20	2473 ± 20			2633 ± 20	[13]
		2443 ± 30	2473 ± 30			2643 ± 30	[14]
		2443 ± 20					[25]
		2443 ± 10	2481 ± 10				[26]
Tb_2O_3	1823			2433			[3]
	1773						[12]
				2448 ± 20		2613 ± 20	[13]
				2448 ± 30		2643 ± 30	[14]
	2148						[27]
	1873						[28]

Table 2.1 (*continued*)

Oxide	Transition						Ref.
	C ⟺ B	B ⟺ A	A ⟺ H	B ⟺ H	C ⟺ H	H ⟺ X	
	2123						[29]
						2638	[30]
Dy_2O_3	2223			2473			[3]
	2133			2453		2650	[12]
	2223 ± 20			2463 ± 20		2618 ± 20	[13]
	2253 ± 30			2463 ± 30		2653 ± 30	[14]
	2423						[27]
	2413						[29]
	2353			2423			[22]
						2648	[30]
Ho_2O_3	2453			2513			[3]
	2453			2453			[12]
	2463 ± 20			2493 ± 20			[13]
	2463 ± 30			2473 ± 30			[14]
Er_2O_3					2553		[3]
					2523		[12]
					2593 ± 20		[13]
					2593 ± 30		[14]
Tm_2O_3					2623		[12]
					2623 ± 20		[13]
					2623 ± 30		[14]
Yb_2O_3					2653		[3]
					2653 ± 20		[13]
					2653 ± 30		[14]

[a] In these works, the high-temperature modification is reported to be of cubic fluorite type.

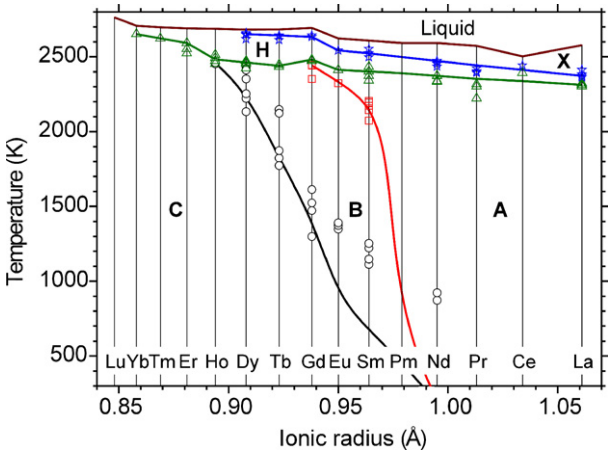

Fig. 2.1. The scheme of polymorphic transformations in lanthanide sesquioxides. Lines connect the assessed temperatures, various symbols represent the experimental values, as reported in the literature. Ionic radii are taken from Ref. [15] for six-fold coordination.

Table 2.2
Assessed temperatures of the reversible solid $\Longleftrightarrow$ solid phase transitions in rare earth sesquioxides at ambient pressure ($P = 1$ bar)

Oxide	Transition	T (K)	Category[a]
Sm_2O_3	C $\Longleftrightarrow$ B	676 ± 20	E
Eu_2O_3	C $\Longleftrightarrow$ B	894 ± 20	E
Gd_2O_3	C $\Longleftrightarrow$ B	1425 ± 20	E
Tb_2O_3	C $\Longleftrightarrow$ B	1823 ± 50	P
Dy_2O_3	C $\Longleftrightarrow$ B	2263 ± 30	P
Ho_2O_3	C $\Longleftrightarrow$ B	2458 ± 20	R
Sm_2O_3	B $\Longleftrightarrow$ A	2173 ± 30	R
Eu_2O_3	B $\Longleftrightarrow$ A	2323 ± 20	R
Gd_2O_3	B $\Longleftrightarrow$ A	2443 ± 10	R
La_2O_3	A $\Longleftrightarrow$ H	2313 ± 20	R
Ce_2O_3	A $\Longleftrightarrow$ H	2338 ± 60	E
Pr_2O_3	A $\Longleftrightarrow$ H	2353 ± 50	E
Nd_2O_3	A $\Longleftrightarrow$ H	2373 ± 30	R
Sm_2O_3	A $\Longleftrightarrow$ H	2403 ± 30	R
Eu_2O_3	A $\Longleftrightarrow$ H	2413 ± 20	R
Gd_2O_3	A $\Longleftrightarrow$ H	2481 ± 10	R
Tb_2O_3	B $\Longleftrightarrow$ H	2440 ± 20	R
Dy_2O_3	B $\Longleftrightarrow$ H	2463 ± 20	R
Ho_2O_3	B $\Longleftrightarrow$ H	2483 ± 30	R
Y_2O_3	C $\Longleftrightarrow$ H	2600 ± 30	R
Er_2O_3	C $\Longleftrightarrow$ H	2593 ± 50	P
Tm_2O_3	C $\Longleftrightarrow$ H	2623 ± 20	R
Yb_2O_3	C $\Longleftrightarrow$ H	2653 ± 20	R
La_2O_3	H $\Longleftrightarrow$ X	2373 ± 30	R
Ce_2O_3	H $\Longleftrightarrow$ X	2413 ± 30	E
Pr_2O_3	H $\Longleftrightarrow$ X	2443 ± 30	E
Nd_2O_3	H $\Longleftrightarrow$ X	2473 ± 10	R
Sm_2O_3	H $\Longleftrightarrow$ X	2523 ± 30	R
Eu_2O_3	H $\Longleftrightarrow$ X	2543 ± 20	R
Gd_2O_3	H $\Longleftrightarrow$ X	2633 ± 20	R
Tb_2O_3	H $\Longleftrightarrow$ X	2643 ± 30	P
Dy_2O_3	H $\Longleftrightarrow$ X	2653 ± 40	P

[a] R – recommended values, P – probable values, E – estimated values.

lacking or very uncertain. Owing to the regular dependence of the transition temperatures on ionic radius, trends can easily be followed and the possibility for a systematic error is significantly reduced. Here and in the following the term "ionic radius" is understood as the radius of trivalent cation in six-fold coordination as reported by Shannon and Prewitt [15]. Additional measurements on single sesquioxides found in the literature [16–30] are used to eliminate the possible misprints in the publications and to justify the selection.

2.1.1. The C $\Longleftrightarrow$ B phase transition

The C $\Longleftrightarrow$ B phase transition is confirmed to occur reversibly in Tb_2O_3, Dy_2O_3, and Ho_2O_3, but only for the latter experimentally measured values are in good agreement, so that the mean value is recommended. The probable values for Tb_2O_3 and Dy_2O_3 are

selected based on analysis of the temperature dependence of transition pressure (see Section 5.2). The temperatures of C $\Longleftrightarrow$ B phase transitions in Sm_2O_3, Eu_2O_3, and Gd_2O_3 are estimated from the Y_2O_3–Sm_2O_3, Y_2O_3–Eu_2O_3, and Y_2O_3–Gd_2O_3 phase diagrams, respectively (see Section 7). Careful extrapolations of the C + B two-phase fields in these systems does show that C-Sm_2O_3, C-Eu_2O_3, and C-Gd_2O_3 are thermodynamically stable phases at room temperature and hence, the corresponding C $\Longleftrightarrow$ B phase transitions are reversible.

2.1.2. The B $\Longleftrightarrow$ A phase transition

The reported transformations temperatures in Sm_2O_3, Eu_2O_3, and Gd_2O_3 are in good accord. The only exception is the work of Coutures et al. [12], where the reported temperatures are 90–100 K lower than the mean value of the other results. The recommended values are weighted averages of three measurements [3,13,14].

2.1.3. The A $\Longleftrightarrow$ H phase transition

All measurements of the transition temperatures in La_2O_3, Nd_2O_3, Sm_2O_3, Eu_2O_3, and Gd_2O_3 are in good agreement. The recommended values in Table 2.2 correspond to the weighted averages of four determinations [3,12–14], except for Gd_2O_3, where the transition temperature is taken from the more precise measurements by Barkhatov et al. [26]. For Ce_2O_3, two coinciding values have been published, both by the same group [13,14]. In the case of Pr_2O_3, the measurements have been carried out by two groups, [3,13,14] and there is a large discrepancy between their results. If a possible undercooling effect is taken into account, the values given in Refs. [13,14] appear to be more reliable. The selected values are estimated in this work by linear interpolation between La_2O_3 and Nd_2O_3. Note that the corresponding transition temperatures for Sm_2O_3 and Eu_2O_3 lie on the same straight line. The value thus obtained for Pr_2O_3 is also consistent with an estimate from high-temperature X-ray diffraction experiments [3]. A large uncertainty of the measurements might have been caused by the variations of oxygen stoichiometry of the sample, since the oxidation states +2 and +4 are relatively stable for cerium and praseodymium. A careful redetermination of the A $\Longleftrightarrow$ H transition temperature in Ce_2O_3 and Pr_2O_3 under controlled partial pressure of oxygen would be desirable.

2.1.4. The B $\Longleftrightarrow$ H phase transition

The reported values for the temperature of the B $\Longleftrightarrow$ H transformation in Tb_2O_3, Dy_2O_3 and Ho_2O_3 [3,12–14], except for the lower value from Ref. [22] show good consistency. The recommended values are taken as unweighted averages of those measurements.

2.1.5. The C $\Longleftrightarrow$ H phase transition

Three values for the C $\Longleftrightarrow$ H phase transition in Y_2O_3 [3,12,13] are in good agreement. The difference of 50 K is attributed to the possible overheating/undercooling effects. The highest temperature (2663 K) reported in Ref. [14] seems to be a misprint. In the course of studies of R_2O_3–Y_2O_3 phase diagrams by the same group [31–34] this transformation is found to occur at 2643 K. The recommended value (2600 ± 30 K) is accepted from the work of Shpil'rain and co-workers [16], since it represents the most accurate determination by enthalpy and electrical conductivity measurements and is also close to the mean value of the other reports. Some studies claimed that the high-temperature phase of Y_2O_3 is of cubic fluorite type [20,35]. This is however, clearly at variance with the

existence of continuous solid solutions of H-type in several R_2O_3–Y_2O_3 systems (see Section 4). No discrepancies in the reported transition temperatures for Tm_2O_3 and Yb_2O_3 [3,12–14] are found and the corresponding values are thus recommended. Owing to the possible undercooling effect in the thermal analysis method [3,12] the probable value for Er_2O_3 is weighted towards DTA measurements [13,14]. This selection also gives rise to a reasonable linear dependence of the C $\Longleftrightarrow$ H transition temperature on ionic radius (Fig. 2.1).

2.1.6. The H $\Longleftrightarrow$ X phase transition

All measurements of the transition temperatures in La_2O_3, Nd_2O_3, Sm_2O_3, Eu_2O_3, and Gd_2O_3 are in good accord. The recommended values in Table 2.2 correspond to the weighted averages of four determinations [3,12–14]. For Ce_2O_3 and Pr_2O_3 the experimental values show the same trend as in the case of A $\Longleftrightarrow$ H transition (see above) and therefore the selected values are estimated through the linear interpolation. Among the reported temperatures for Tb_2O_3 and Dy_2O_3, which are in fair agreement the highest measured values are selected. It can be justified by the fact, that all the H $\Longleftrightarrow$ X transition temperatures in lanthanide sesquioxides increase with decreasing ionic radius (Fig. 2.1).

Table 2.3
Reported transition pressures for the polymorphic transformations in rare earth sesquioxides at different temperatures

Oxide	Transition	T (K)	P (GPa)	Ref.
Sc_2O_3	C ⇒ B	1273	8–11	[38]
	C ⇒ A	~298	40.9 ± 1.3	[48]
Y_2O_3	C ⇒ B	1273	2.1	[36]
	C ⇒ B	~298	12	[47]
	C ⇒ B	~298	10 ± 2	[44]
	C ⇒ B	~298	12.8	[45]
	B ⇒ A	~298	19 ± 2	[44]
	B ⇒ A	~298	21.8	[45]
Sm_2O_3	B ⇒ A	~298	3.2–3.9	[39]
Eu_2O_3	C ⇒ B	~298	8	[40]
	B ⇒ A	~298	4.3 ± 0.4	[42]
Tb_2O_3	C ⇒ B	1273	0.6	[36]
Dy_2O_3	C ⇒ B	1273	1.0	[36]
		492	1.81	[37]
		589	1.72	[37]
		686	1.63	[37]
		783	1.55	[37]
		839	1.49	[37]
		885	1.44	[37]
		1018	1.32	[37]
		1380	0.99	[37]
Ho_2O_3	C ⇒ B	1273	1.5	[36]
Er_2O_3	C ⇒ B	1273	2.0	[36]
Tm_2O_3	C ⇒ B	1273	2.5	[36]
Yb_2O_3	C ⇒ B	1273	3.0	[36]
	C ⇒ B	~298	13 ± 1	[43]
Lu_2O_3	C ⇒ B	1273	3.6	[36]

The data of Hoekstra [36] are shown for 1273 K only, while the corresponding transition pressures at 823 and 1723 K are 0.5 GPa higher and lower values, respectively.

2.2. Transitions at elevated pressures

Since C ⇒ B and B ⇒ A phase transitions in rare earth sesquioxides are accompanied by the volume contraction (see Section 3.1) the application of pressure increases the stability of B- and A-structures relative to C- and B-modifications, respectively. Hoekstra [36] determined the equilibrium pressures for the C ⇒ B transformation in R_2O_3 (R = Y, Eu–Lu) at 823, 1273 and 1723 K. The $P - T$ equilibrium boundary separating B- and C-Dy_2O_3 has been studied by Seck an co-workers [37]. Reid and Ringwood [38] reported that cubic Sc_2O_3 transforms into the monoclinic B-modification between 8 and 11 GPa at 1273 K. At very high pressures, the polymorphic C ⇒ B ⇒ A transitions are known to occur even around the room temperature under static load [39–45] or shock compression [46–48]. The available literature data on the transition pressures are compiled in Table 2.3. It can be seen that the data of different works concerning the pressure-induced C ⇒ B transition in Y_2O_3 and Dy_2O_3 and B ⇒ A transition in Y_2O_3 are in good agreement.

3. Thermodynamic data for solid and liquid phases

3.1. The molar volume

The molar volume of A-, B-, and C-type structures of rare earth sesquioxides (per R_2O_3 formula unit) is shown in Fig. 3.1 as a function of ionic radius. The room temperature data for A-R_2O_3 (R = La–Pm), B-R_2O_3 (R = Pm–Lu), and C-R_2O_3 (R = Pr–Lu) are taken from Ref. [2] except for B-Tb_2O_3, B-Dy_2O_3, and B-Ho_2O_3, which are from Ref. [49], while the values for C-Sc_2O_3 and C-Y_2O_3 are accepted from Refs. [50,51]. In addition, the molar volume of A-Sc_2O_3, B-Sc_2O_3, and B-Y_2O_3 is taken from the publications, where the corresponding phases have been obtained at elevated pressures and characterized at normal conditions [36,38,48]. For the phase transition B ⇒ A in Eu_2O_3, the volume change of

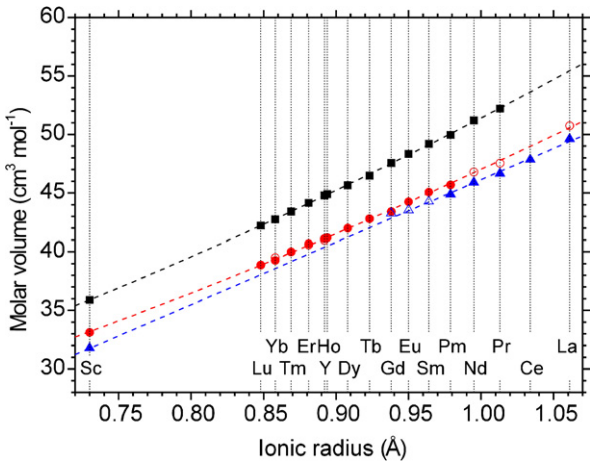

Fig. 3.1. The molar volume per R_2O_3 formula unit of A-type (triangles), B-type (circles), and C-type (squares) rare earth sesquioxides plotted against ionic radius. Filled symbols: measured values, open symbols: extrapolated values. The dashed lines represent the least-squares fit to the second-order polynomial function.

−1.6% at a pressure of 4.3 GPa is reported [42]. Strictly speaking, this value is not equal to the volume change at ambient pressure, but the necessary correction can be neglected in the first approximation. Thus, the molar volume of A-Eu_2O_3 is calculated and included in Fig. 3.1. The molar volumes of B-Y_2O_3, B-La_2O_3, B-Pr_2O_3, B-Nd_2O_3, A-Sm_2O_3, A-Gd_2O_3, B-Er_2O_3, and B-Yb_2O_3 are calculated according to the Vegard's law extrapolating the composition dependence of the unit-cell volume of solid solutions R'_2O_3–R''_2O_3 [12,32,49,52–55] onto 100% R'_2O_3 and/or R''_2O_3 (see also Section 4). It can be seen that the extrapolated values are in good agreement with those measured directly and follow the same trend: the dependencies of the molar volume on ionic radius are almost linear (slightly parabolic) across the entire rare earth series, while the relative volume changes at C ⇒ B and B ⇒ A phase transitions are about −8% and −2%, respectively. The assessed values of the molar volume of A-, B-, and C-structures of the rare earth sesquioxides at room temperature are compiled in Table 3.1.

In Fig. 3.2 the absolute values of the volume change in the course of C ⇒ B and B ⇒ A phase transitions are plotted versus ionic radius. As a general trend, the volume difference increases monotonically with increasing radius of cation. An exception is a hypothetical B ⇒ A transition in Sc_2O_3, for which the absolute value of volume change exceeds the corresponding values for other rare earth sesquioxides. An interesting feature is a singularity at Gd_2O_3, which is clearly seen as a local maximum and a local minimum (actually, the volume change almost drops to zero) on the curves corresponding to C ⇒ B and B ⇒ A transformations, respectively. The anomalous behaviour of gadolinium sesquioxide seems to originate from the electronic configuration of Gd^{+3}-ion, which has a half-filled f-shell.

The molar volumes of the H- and X-phases have only been measured at high temperatures. The absolute values cannot be compared directly as the transition temperatures are different. The A ⇒ H transition is accompanied by the lattice expansion, while the H ⇒ X

Table 3.1
Assessed molar volumes of A-, B- and C-type rare earth sesquioxides ($cm^3\ mol^{-1}$) at room temperature

Oxide	Structure		
	A-type	B-type	C-type
Sc_2O_3	31.781	33.129	35.911
Y_2O_3	(40.385)	41.132	44.804
La_2O_3	49.627	50.748	(55.442)
Ce_2O_3	47.842	(48.985)	(53.637)
Pr_2O_3	46.675	47.545	52.202
Nd_2O_3	45.893	46.796	51.198
Pm_2O_3	44.886	45.697	49.960
Sm_2O_3	44.263	45.082	49.200
Eu_2O_3	43.538	44.246	48.328
Gd_2O_3	43.265	43.403	47.574
Tb_2O_3	(42.036)	42.818	46.473
Dy_2O_3	(41.237)	42.004	45.654
Ho_2O_3	(40.491)	41.204	44.910
Er_2O_3	(39.799)	40.677	44.163
Tm_2O_3	(39.160)	39.994	43.405
Yb_2O_3	(38.574)	39.266	42.757
Lu_2O_3	(38.041)	38.878	42.225

Values given in parentheses are estimated by interpolation.

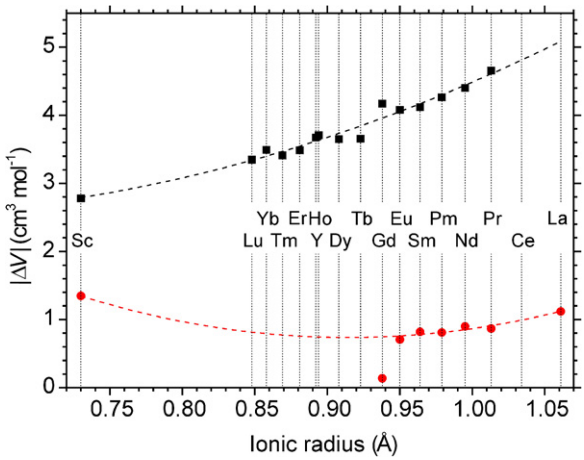

Fig. 3.2. The absolute values of the volume change in the course of C ⇒ B (squares) and B ⇒ A (circles) phase transitions in rare earth sesquioxides versus ionic radius. Lines represent the estimated general trends.

transition results in the lattice contraction [3]. The average relative volume changes upon these phase transitions in La_2O_3, Pr_2O_3, Nd_2O_3, Sm_2O_3, Gd_2O_3, Tb_2O_3 and Dy_2O_3 according to the *in situ* high-temperature X-ray diffraction experiments [3] are +3% and −0.5%, respectively.

The melting of rare earth sesquioxides is accompanied by the significant increase of the molar volume. The relative variation of volume at fusion of Y_2O_3, La_2O_3, Nd_2O_3, Sm_2O_3, Eu_2O_3, Gd_2O_3, Er_2O_3 and Yb_2O_3 has been calculated by Granier and Heurtault [56] from the density measurements between the melting point and ~3200 K. In Fig. 3.3, the values

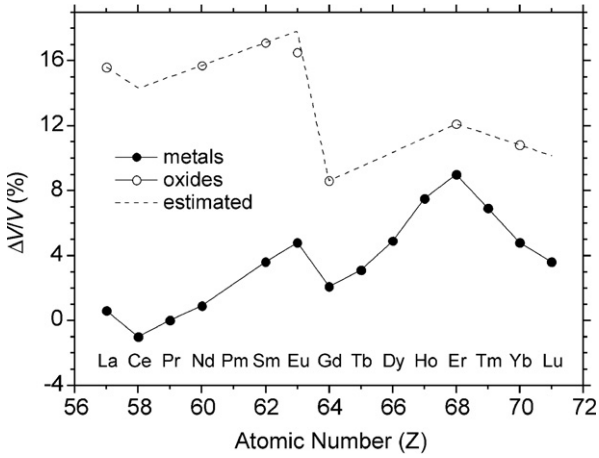

Fig. 3.3. The relative change of the molar volume of rare earth sesquioxides and metals upon melting as function of atomic number of the rare earth element. The dashed line shows the trend estimated by linear interpolation. The solid line is only guide for eye.

of the relative volume change on melting of the rare earth sesquioxides along with the corresponding data for the rare earth metals [57,58] are plotted versus atomic number of the element. It can be seen that oxides and metals show similar behaviour. This observation is used to estimate the volume change at fusion of other lanthanide sesquioxides by linear interpolation (dashed line in Fig. 3.3) as well as the entropy of melting (see Section 5.5). The numerical values of the relative volume change are given in Table 3.2.

3.2. Enthalpies, entropies and heat capacities

The measurements of enthalpy of formation of C-Sc_2O_3 and C-Y_2O_3 available in the literature were critically assessed in Refs. [59,60]. The standard enthalpies of formation of A-type R_2O_3 (R = La–Nd), B-type R_2O_3 (R = Sm–Gd), and C-type R_2O_3 (R = Tb–Lu) determined calorimetrically were reviewed by Cordfunke and Konings [6]. The value for A-Pm_2O_3 has been estimated [6].

The enthalpies of formation of C-type Pr_2O_3, Nd_2O_3, Sm_2O_3 and Eu_2O_3 have been measured directly by solution calorimetry [61–64]. In addition, enthalpy of C ⇒ B phase transition in Gd_2O_3 has been measured by Kolitsch [65] by differential scanning calorimetry, while the corresponding enthalpies of B ⇒ A and A ⇒ H transformations have been determined by Barkhatov and co-workers with the use of an adiabatic boiling calorimeter [26]. The same technique has been employed by Shpil'rain et al. [16,66] to measure the enthalpy of C ⇒ H phase transition in Y_2O_3 as well as the heat of fusion of C-Sc_2O_3 and H-Y_2O_3. Recently, the enthalpy of a solid-state phase transition and the enthalpy of fusion in Y_2O_3 has been remeasured using high-temperature DTA [21]. Atou et al. [48] estimated the enthalpy of C ⇒ A phase transition in Sc_2O_3 from the analysis of shock compression curve. These values are discussed in Section 5.3.

Table 3.2
Assessed relative variation of molar volume at fusion of rare earth sesquioxides [56]

Oxide	T_m (K)	Structure type at T_m	$\Delta V/V$ (%)
Y_2O_3	2712	H	11.2
La_2O_3	2578	X	15.6
Ce_2O_3	2503	X	(14.3)
Pr_2O_3	2573	X	(15.0)
Nd_2O_3	2593	X	15.7
Pm_2O_3	2593	X	(16.4)
Sm_2O_3	2608	X	17.1
Eu_2O_3	2623	X	16.5
Gd_2O_3	2693	X	8.6
Tb_2O_3	2683	X	(9.5)
Dy_2O_3	2681	X	(10.4)
Ho_2O_3	2688	H	(11.2)
Er_2O_3	2691	H	12.1
Tm_2O_3	2698	H	(11.4)
Yb_2O_3	2708	H	10.8
Lu_2O_3	2763	C	(10.2)

The melting points (T_m) are from Refs. [10,11]. Values given in parentheses are estimated by interpolation.

Low-temperature heat capacity data, which permit the calculation of the standard entropy ($^{\circ}S_{298}$) are available for all rare earth sesquioxides, except for Pm_2O_3 and Tb_2O_3 [67–77]. The reported values refer to A-R_2O_3 for R = La–Nd, B-R_2O_3 for R = Sm, and C-R_2O_3 for R = Gd, Dy–Lu). For Eu_2O_3, the heat capacity of both C- and B-modifications has been measured between 8 and 300 K and the enthalpy of C ⇒ B phase transition is reported [76]. Several datasets, which are available for Y_2O_3, Ce_2O_3, La_2O_3, and Nd_2O_3 have been compared in the critical assessment works [7,60].

Except for Pm_2O_3, heat capacities and/or enthalpy increments above room temperature have been measured for all rare earth sesquioxides in the polymorphic modification, which is stable at normal conditions, while the data for both C- and B-structures of Sm_2O_3, Eu_2O_3, and Gd_2O_3 are available [16,65,66,68,77–98]. The enthalpy increments for the liquid Y_2O_3 and Sc_2O_3 are also reported [16,66]. These data are analysed in Section 6.

4. The $R'_2O_3 - R''_2O_3$ systems

A number of investigations have been devoted to the quasibinary systems R'_2O_3–R''_2O_3, where R′ and R″ are different rare earth elements. Early studies by Schneider and Roth [99] revealed the existence of extended A- and C-type solid solutions in the subsolidus region, while the B-type structures have been found to occur in the intermediate range of compositions, when mixing A-R'_2O_3 and $C - R''_2O_3$. Thus, the substitution of a smaller cation by a larger one acts similarly to the increase of external pressure and the sequence of transformations with increasing ionic radius is C ⇒ B ⇒ A. The monotonic shift of the phase boundaries, when R′ and R″ are systematically varied allowed the authors [99] to construct a generalized diagram showing the stability of A-, B-, and C-structures as the plot of the average ionic radius versus mole percent of the smaller cation. In addition, the perovskite-type compounds $LaRO_3$ have been found to exist as discrete phases for R = Er, Tm, Yb, and Lu. The homogeneity ranges of the A-, B-, and C-type solid solutions at different temperatures have been studied in the systems La_2O_3–R_2O_3 (R = Y, Ce–Lu) [49,52,100] and Y_2O_3–R_2O_3 (R = La–Lu) [100]. The results have shown an acceptable agreement with previous work [99]. In addition, perovskite-type compound $LaHoO_3$ has been observed [100].

Complete phase diagrams are reported for the systems La_2O_3–R_2O_3, R = Y [31,101], R = Nd, Sm, Gd, Dy–Yb [12], Pr_2O_3–Dy_2O_3 [53], Nd_2O_3–Y_2O_3 [17,54,55], Nd_2O_3–Yb_2O_3 [55],Y_2O_3–Sm_2O_3 [32], Y_2O_3–Gd_2O_3 [32], Y_2O_3–Eu_2O_3 [19], Y_2O_3–Dy_2O_3 [33], Y_2O_3–Tb_2O_3 [18], Y_2O_3–Er_2O_3 [34], and Sc_2O_3–R_2O_3, R = Y, La, Nd, Sm, Gd, Dy–Lu [102]. The formation of the perovskite-type phase $LaRO_3$, which also has a homogeneity range has been confirmed for R = Tm, Er, Yb, and Y [12], whereas isostructural compound $RScO_3$ has been found to exist in the systems with R = La–Ho [102]. Additional intermediate phase of composition R_3ScO_6 (Mg_3TeO_6-type) has been observed in the case of R = Dy, Ho and Y [102]. The complete miscibility of the rare earth sesquioxides in the liquid state as well as in the high-temperature H- and X-phases has been confirmed in these works. The close chemical similarity of the rare earth elements is often reflected in the very narrow two-phase fields on R'_2O_3–R''_2O_3 phase diagrams, which are difficult to detect experimentally. Enthalpy of mixing in the C-type solid solution Nd_2O_3–Y_2O_3 has been determined by Tikhonov and co-workers [103]. Positive and negative deviations from ideality have been observed in the Y_2O_3-rich and Nd_2O_3-rich regions, respectively.

5. Estimation of missing data

5.1. The relationship between pressure and temperature of a phase transition

The available calorimetric data on the energetics of the polymorphic phase transitions in rare earth sesquioxides are scarce (see Section 3.2). However, the enthalpy and entropy of a phase transition can be estimated from the dependence of the transition pressure on temperature using the data reviewed in Section 2.2 and standard thermodynamic relations, provided that the corresponding volume change is known. The Gibbs energy change due to the phase transition is given by

$$\Delta G(P,T) = \Delta^\circ H(T) - T\Delta^\circ S(T) + \int_0^P \Delta V(P,T)\,\mathrm{d}P, \tag{5.1}$$

where $\Delta^\circ H\,(T)$, $\Delta^\circ S(T)$, and $\Delta V(P,T)$ are the enthalpy, the entropy of phase transition at zero pressure, and the corresponding volume change, respectively. In equilibrium, $\Delta G(P,T) = 0$ and

$$\Delta^\circ S(T) = \frac{\Delta^\circ H(T)}{T} + \frac{1}{T}\int_0^P \Delta V(P,T)\,\mathrm{d}P \tag{5.2}$$

In order to integrate Eq. (5.2) the temperature and pressure dependence of the volume change must be known. However, there is a lack of the such data for rare earth sesquioxides. Assuming that both phases have the same compressibility and thermal expansion, i.e., $\Delta V = \text{const.}$, Eq. (5.2) can be transformed to

$$\Delta^\circ S(T) = \frac{\Delta^\circ H(T) + P\Delta V}{T} \tag{5.3}$$

The temperature dependence of $\Delta^\circ H$ and $\Delta^\circ S$ originates from the differences in heat capacity of two polymorphic modifications, which is not known, except for C- and B-forms of Sm_2O_3, Eu_2O_3, and Gd_2O_3. Thus, Eq. (5.3) may be further simplified assuming that a phase transition has no effect on the heat capacity, i.e.,

$$\Delta^\circ S = \frac{\Delta^\circ H}{T} + \frac{P\Delta V}{T} \tag{5.4}$$

Eq. (5.4) can be used to calculate the transition temperature as a function of pressure and *vice versa*

$$T_{\text{trans}} = \frac{\Delta^\circ H}{\Delta^\circ S} + \frac{\Delta V}{\Delta^\circ S}P \tag{5.5}$$

$$P_{\text{trans}} = -\frac{\Delta^\circ H}{\Delta V} + \frac{\Delta^\circ S}{\Delta V}T \tag{5.6}$$

By differentiation of Eq. (5.6) one obtains the Clausius–Clapeiron relation

$$\frac{\mathrm{d}P}{\mathrm{d}T} = \frac{\Delta S}{\Delta V} \tag{5.7}$$

In Fig 5.1 the C $\Longleftrightarrow$ B and B $\Longleftrightarrow$ A transition pressures for rare earth sesquioxides are plotted versus temperature. The observed straight-line dependencies for C $\Longleftrightarrow$ B transitions confirm the validity of Eq. (5.6). The only exceptions are very high C $\Rightarrow$ B transition

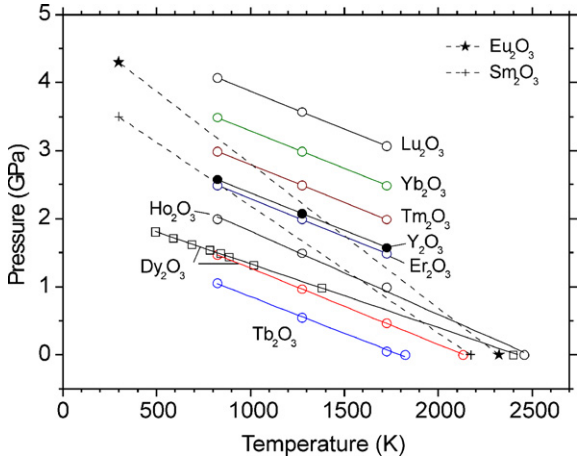

Fig. 5.1. Equilibrium pressures at which the polymorphic transformations in rare earth sesquioxides have been observed plotted versus temperature. Circles and squares refer to C $\Longleftrightarrow$ B transition data from Refs. [36,37], respectively. The data for Eu_2O_3 and Sm_2O_3 refer to B $\Longleftrightarrow$ A transition [39,42].

pressures at room temperature reported for Y_2O_3, Eu_2O_3 and Yb_2O_3 (not shown in the Figure, see Table 2.3). On the one hand, those values are likely overpressures. On the other hand, these pressure-induced phase transitions are reported to occur irreversibly [40,43–45] so that using equilibrium thermodynamics for analysis of such data may not be appropriate. In contrast, C $\Longleftrightarrow$ B phase transitions at higher temperatures as well as B $\Longleftrightarrow$ A phase transitions at room temperature have been confirmed to be fully reversible.

In the case of Ho_2O_3, extrapolation to the 1 bar pressure gives the transition temperature, which is well-consistent with the selected value in Table 2.2. For Tb_2O_3, the straight line is obtained with the lowest measured transition temperature at normal pressure, 1823 K [3], which is selected as a probable value. Two datasets are available for Dy_2O_3 [36,37]. Extrapolation to the 1 bar pressure gives rise to the values of 2145 and 2445 K, respectively, which approximately correspond to the boundaries of experimental scatter (Table 2.1). The selected value is therefore taken as a weighted average of seven measurements. Fitting the data in Fig. 5.1 to Eq. (5.6) and taking the volume change from Table 3.1 the enthalpies and entropies of the corresponding phase transitions are derived. The reliability of thus obtained thermodynamic data depends, of course, on the accuracy in the measurements of transition pressures.

5.2. The energetics of C $\Rightarrow$ B phase transition

A combined analysis of the enthalpies and entropies obtained via Eq. (5.6) and the available literature data allows to establish the general trends in the energetics of polymorphic transformations in rare earth sesquioxides. All these values are compiled in Table 5.1. In the case of Eu_2O_3, the transition enthalpy given by Lyutsareva and co-workers [76] is preferred over the value measured by Fitzgibbon et al. [64], which is unusually high compared to Gd_2O_3 and Sm_2O_3, while its uncertainty is quite large. By plotting enthalpies of C $\Rightarrow$ B transition in R_2O_3 (R = Tb–Lu) versus ionic radius a straight line is obtained (Fig. 5.2). Extrapolating this dependence allows to estimate the enthalpy of C $\Rightarrow$ B

Table 5.1
Thermodynamic data for the polymorphic phase transitions in rare earth sesquioxides at ambient pressure, $P = 1$ bar

Oxide	Transition	ΔH (kJ mol^{-1})	ΔS (J mol^{-1} K^{-1})	Category[a]	Reference	Comments
Sc_2O_3	C ⇒ A	50.0	–	D	[48]	
Y_2O_3	C ⇒ B	12.833	4.080	D	[36]	
	C ⇒ H	25.1 ± 8.5	9.654	C	[16]	At 2600 K
	C ⇒ H	47.7 ± 3.0	18.500	C	[21]	At 2581 K
Pr_2O_3	C ⇒ A	0.0 ± 0.3	–	C	[61]	At 298.15 K
Nd_2O_3	C ⇒ A	24.5 ± 4.2	–	C	[62]	At 298.15 K
Sm_2O_3	C ⇒ B	3.8 ± 2.6	–	C	[63]	At 298.15 K
	B ⇒ A	3.324	1.529	D	[39]	
Eu_2O_3	C ⇒ B	12.1 ± 6.0	–	C	[64]	At 298.15 K
	C ⇒ B	4.800	–	C	[76]	At 1348 K
	B ⇒ A	3.494	1.504	D	[42]	
Gd_2O_3	C ⇒ B	5.3 ± 0.4	–	C	[65]	At 1561 K
	B ⇒ A	6.3 ± 3.0	2.579	C	[26]	At 2443 K
	A ⇒ H	34.7 ± 3.0	13.986	C	[26]	At 2481 K
Tb_2O_3	C ⇒ B	7.054	3.925	D	[36]	
Dy_2O_3	C ⇒ B	8.700	4.056	D	[36]	
	C ⇒ B	8.275	3.384	D	[37]	
Ho_2O_3	C ⇒ B	11.253	4.522	D	[36]	
Er_2O_3	C ⇒ B	11.880	3.873	D	[36]	
Tm_2O_3	C ⇒ B	13.326	3.790	D	[36]	
Yb_2O_3	C ⇒ B	15.381	3.879	D	[36]	
Lu_2O_3	C ⇒ B	16.691	3.719	D	[36]	

[a] C – calorimetric value, D – derived from the pressure–temperature dependence.

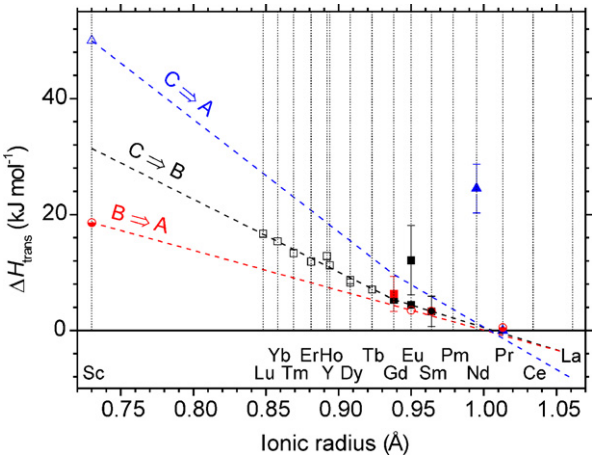

Fig. 5.2. Enthalpies of C ⇒ B (squares), B ⇒ A (circles) and C ⇒ A (triangles) phase transitions in rare earth sesquioxides plotted against ionic radius. Filled symbols: values measured experimentally, open symbols: values derived from the pressure–temperature dependencies, half-filled symbols: enthalpies of B ⇒ A transition calculated from the available enthalpies of C ⇒ A transitions by subtracting the extrapolated enthalpies of C ⇒ B transitions. Lines represent the estimated general trends.

transformation in Sc_2O_3, which is then 31.378 kJ mol^{-1}. Another straight line is connecting the enthalpies of C ⇒ B transition in Gd_2O_3, Eu_2O_3, and Sm_2O_3 at the transition temperatures (filled squares in Fig. 5.2), where the corresponding values, 5.325, 4.395 and 3.312 kJ mol^{-1}, respectively are calculated using the data from Table 5.1 and known enthalpy increments of C- and B-modifications of the oxides. Extrapolating this line beyond the ionic radius of Sm^{+3}, gives the estimate for the enthalpies of C ⇒ B transition in Pm_2O_3, Nd_2O_3, Pr_2O_3, Ce_2O_3, and La_2O_3. The two lines join on gadolinium sesquioxide thus indicating a remarkable consistency between measured enthalpy values and those derived from the pressure–temperature dependencies.

5.3. The energetics of B ⇒ A and C ⇒ A phase transitions

Enthalpy of B ⇒ A transition in Gd_2O_3 has been measured directly, while those in Eu_2O_3 and Sm_2O_3 are derived from the pressure–temperature dependencies (Table 5.1). Besides, enthalpies of B ⇒ A transitions in Sc_2O_3 and Pr_2O_3 are calculated from the available enthalpies of C ⇒ A transitions by subtracting the extrapolated enthalpies of C ⇒ B transitions (the reported enthalpy of C ⇒ A phase transition in Nd_2O_3 [62] is exceptionally high and obviously inconsistent with the general trends). The additivity of enthalpies implies that all these phase transformations, i.e., C ⟺ B ⟺ A are to be considered as reversible. The five values apparently lie on the straight line (Fig. 5.2).

The entropies of C ⇒ B and B ⇒ A transitions show weak dependencies on the radius of cation (Fig. 5.3). Referring to Fig. 3.2 it is reasonable to assume that the $\Delta S/\Delta V$ ratio holds constant across the whole rare earth series, in accordance with Eq. (5.7). Indeed, the correlated behaviour of the volume change and entropy change as functions of ionic radius is evident and even the singularity at Gd_2O_3 is clearly seen in both cases. It is also clear that the greater change of the molar volume corresponds to the greater change of the entropy. Then, any unknown ΔS value can be calculated from the corresponding volume

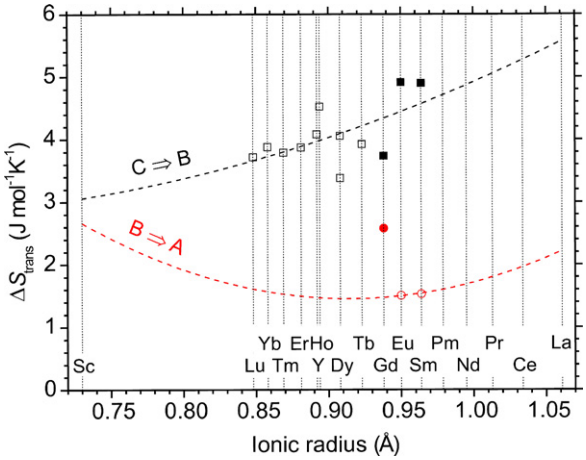

Fig. 5.3. Entropies of C ⇒ B (squares) and B ⇒ A (circles) phase transitions in rare earth sesquioxides plotted against ionic radius. Filled symbols: values obtained from direct experiment (B ⇒ A transition in Gd_2O_3) or estimated from the phase diagrams (C ⇒ B transitions in Sm_2O_3, Eu_2O_3, and Gd_2O_3), open symbols: values derived from the pressure–temperature dependencies (Table 5.1) Lines represent the estimated general trends.

change, which is taken as a function of ionic radius using single coefficient of proportionality for C ⇒ B (1.09583 J cm^{-3} K^{-1}) and B ⇒ A (1.97643 J cm^{-3} K^{-1}) phase transitions. The results of such calculations are shown in Fig. 5.3 as dashed lines, while the sum of two curves (not shown) will represents the entropy of the reversible C ⇒ A phase transition. Finally, the enthalpies of C ⇒ B transformations in Tb_2O_3, Dy_2O_3 and Ho_2O_3 are recalculated using the assessed temperatures of these phase transitions at ambient pressure (Table 2.2).

The established energetic trends are consistent with the scheme of polymorphic transformations in lanthanide sesquioxides at normal pressure (cf. Fig. 2.1). Indeed, the enthalpies of C ⇒ B, B ⇒ A, and C ⇒ A transitions become negative beyond the ionic radius of about 1.00 Å (i.e., between Nd_2O_3 and Pr_2O_3), while the entropies remain always positive (Figs. 5.2 and 5.3). This simply means that the corresponding C- and B-structures are unstable with respect to A-form at any temperature, that is the common observation for La_2O_3, Ce_2O_3, Pr_2O_3, and Nd_2O_3. Moreover, simple calculations correctly predict that C ⇒ B transformation in R_2O_3 is likely to occur between room temperature and meting point for R = Pm - Ho and not for R = Er–Lu, Y, and Sc. Similarly, B ⇒ A transformation will take place only for R = Pm–Gd, whereas in no case the direct (and reversible) C ⇒ A transformation can be expected.

5.4. The energetics of A ⇒ H, B ⇒ H, and C ⇒ H phase transitions

All the rare earth sesquioxides, except Sc_2O_3 and Lu_2O_3 transform into H-modification at sufficiently high temperatures. The positive volume change (expansion) clearly distinguishes these phase transitions from the C ⇒ B ⇒ A sequence, where the volume contraction has been observed. The corresponding thermochemical data are confined to the A ⇒ H transformation in Gd_2O_3 and the C ⇒ H transformation in Y_2O_3 (Table 5.1), while the available data on volume change are very scarce and uncertain. The enthalpy of the solid-state phase transition in Y_2O_3 reported by Shpil'rain et al. [16] is preferred, however, over the more recent DTA measurements, 47.7 kJ mol^{-1} [21]. Firstly, DTA method is known to be generally less accurate compared to the direct calorimetric studies. Secondly, the reliability of such data may be questioned, because of severe aging of the thermocouple, which led to the failure of detector after one to three heating/cooling cycles [21]. To estimate the energetics of the corresponding phase transformation into H-phase in other R_2O_3 compounds, the entropy of A ⇒ H (R = La–Eu) and B ⇒ H (R = Tb–Ho) transitions is assumed to be equal to the entropy of A ⇒ H transformation in Gd_2O_3, whereas the entropy of C ⇒ H transitions in R_2O_3 (R = Sc, Er–Lu) is assumed to be equal to that in Y_2O_3. In fact, the entropy of a phase transformation is mainly determined by the crystal structures of two phases and does not change significantly from one element to another within the rare earth series (cf. Fig. 5.3). The enthalpies of A ⇒ H, B ⇒ H, and C ⇒ H transformation in rare earth sesquioxides R_2O_3 (R = Y, La–Yb), are calculated as products of the entropies and the transition temperatures, respectively. A plot of these values versus ionic radius is shown in Fig. 5.4. It demonstrates linear relationships and emphasizes again the peculiarity of Gd_2O_3, which is consistent with the special electronic configuration of Gd^{+3}-ion (cf. Figs. 2.1, 3.2 and 5.3). Similar conclusion can be drawn with respect to Lu_2O_3, where the electronic configuration of cation is $4f^{14}$ and the melting point is the highest one among the rare earth sesquioxides. The enthalpy of the hypothetical C ⇒ H transition in Lu_2O_3 is thus estimated as 27 kJ mol^{-1}, that is slightly higher

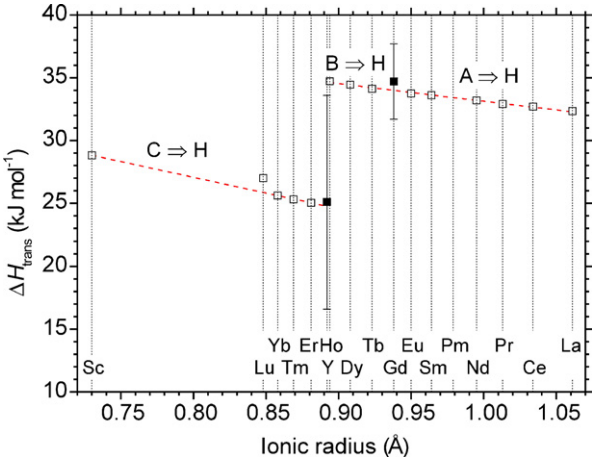

Fig. 5.4. Enthalpies of A ⇒ H, B ⇒ H and C ⇒ H phase transitions in rare earth sesquioxides plotted against ionic radius. Filled symbols: values measured experimentally, open symbols: values accepted in the present work (except for Y_2O_3 and Gd_2O_3). Lines represent the estimated general trends.

than the extrapolated value but gives the transformation temperature well above the melting point.

5.5. The energetics of fusion

The only well-documented measurements of the enthalpy of fusion concerns Sc_2O_3 and Y_2O_3 [16,66], which gave the values of 100.4 ± 8.5 and 83.7 ± 8.5 kJ mol^{-1}, respectively (the more recent data for yttria, 35.6 kJ mol^{-1} [21], do not seem to be very accurate as discussed above). From the melting points of two oxides (2758 and 2712 K) the entropies of fusion are calculated as 36.409 and 30.855 J mol^{-1} K^{-1}, respectively. In "Oxide Handbook" [104] the assessed enthalpy of melting is also given for La_2O_3, Ce_2O_3, Pr_2O_3, Nd_2O_3, and Sm_2O_3. The origin of these data is unclear but they are likely not based on measurements, since neither experimental technique nor the uncertainty are indicated. In the present work, the enthalpies of fusion of the lanthanide sesquioxides are estimated based on the empirical relationship between the entropy of fusion and the corresponding fractional volume change [105], i.e.,

$$\Delta S_{\mathrm{fus}} = nRln2 + \gamma_{\mathrm{G}} C_{\mathrm{V}} (\Delta V / V), \tag{5.8}$$

where n is the number of atoms per formula unit, R is the gas constant, γ_{G} is the Grüneisen parameter, and C_{V} is the heat capacity. Theoretically, it is justified if one assumes that the high temperature solid or liquid states may be described by a set of local strains, which vary randomly both spatially and temporally, but which have well-defined finite mean square values at any fixed coordinate of state [106]. Assuming that the Grüneisen parameter at liquid–solid coexistence does not vary widely from system to system, the entropy of fusion simply scales with the number of atoms for a given fractional change of volume. This prediction has been shown to be approximately obeyed for a number of monovalent, divalent, and trivalent metal halides [107]. In order to apply the relation (5.8) to lanthanide sesquioxides, the available data on the entropy and volume of solid and liquid lanthanide

metals [57,58] are used as a reference. In Fig. 5.5 the values of ΔS_{fus} are plotted against the fractional volume change. A linear fit of the data for metals (solid line) gives $n \approx 1$ as expected for monoatomic systems and the slope $\gamma_G C_V = 0.6538$. The scaling factor to estimate the entropies of fusion of sesquioxides (by which both terms of Eq. (5.8) are to be multiplied) is found to be 2.5 based on the known values for yttria, $\Delta S_{fus} = 30.855$ J mol^{-1} K^{-1} and $\Delta V/V = 11.2\%$ (Table 3.2). Since it is only half of the number of atoms in the R_2O_3 formula unit, one may speculate about the "melting entropy

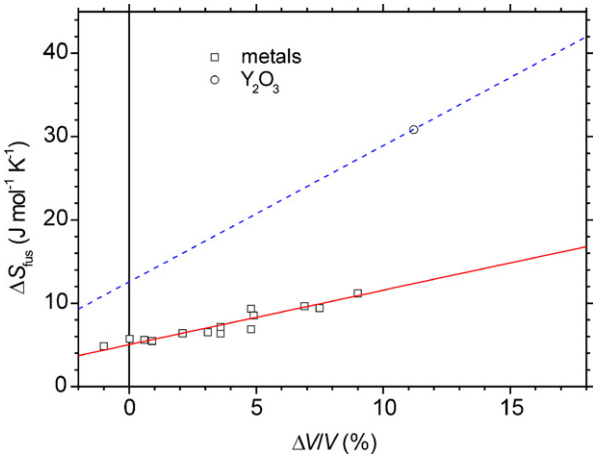

Fig. 5.5. The entropy of fusion of the lanthanide metals and rare earth sesquioxides plotted versus the fractional change of volume upon melting. Solid line: linear fit of the data for metals, dashed line: estimated dependence for rare earth sesquioxides.

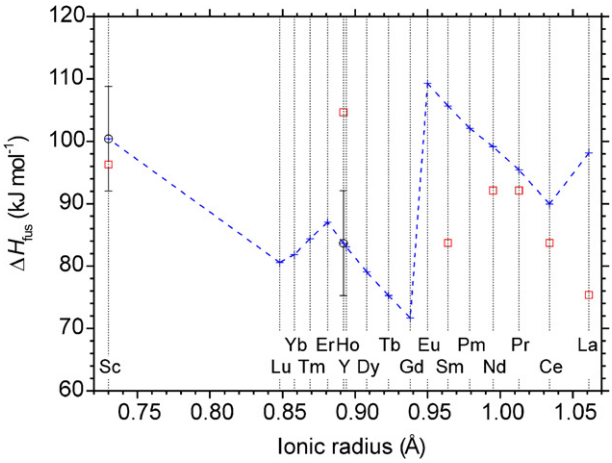

Fig. 5.6. The enthalpy of fusion of rare earth sesquioxides plotted against ionic radius. Circles: experimental data, squares: values from "Oxide Handbook" [104], crosses: values evaluated in the present work. The dashed line is only guide for eye.

deficit" in rare earth sesquioxides. Similar behaviour is known for several groups of halide salts [107], which are superionic conductors (AgI, $SrCl_2$), network-structured compounds ($ZnCl_2$, BeF_2) or molecular crystals ($HgCl_2$). A hypothesis about fast oxygen-ion conductivity in the high-temperature modifications of rare earth sesquioxides seems to be appropriate. Indeed, evidence for fast oxygen diffusion in the H- and X-phases of La_2O_3 has been obtained from neutron diffraction data [108]. In such a case, the anionic sublattice becomes substantially disordered before melting and therefore ΔS_{fus} is lower than predicted by Eq. (5.8). Moreover, a diffraction study of molten Y_2O_3 with anomalous X-ray scattering at 2770 K has shown that the liquid retains the local structure of the high-temperature solid phase [109]. It should be noted that slope of ΔS_{fus} vs. $\Delta V/V$ plot for sesquioxides must be higher than that for metals as the C_V term also scales with the number of atoms. According to the dashed line in Fig. 5.5 the entropy of melting for each R_2O_3 (R = La–Lu) is calculated from the corresponding relative variation of molar volume (Table 3.2), while the enthalpy of fusion is evaluated in turn as a product of ΔS_{fus} and T_m. The results for the enthalpy are shown in Fig. 5.6 together with the available data. As a general trend the enthalpy of fusion increases with decreasing ionic radius with a large drop at gadolinium sesquioxide and local minima at cerium and lutetium sesquioxides. Such dependence is quite consistent with the subdivision of the lanthanide series onto "light" and "heavy" elements as well as with the well-known anomalous behaviour of Ce_2O_3, Gd_2O_3, and Lu_2O_3. It can be seen that the calculated enthalpies of fusion are also not very far from the corresponding values given in "Oxide Handbook" [104], although their reliability may be questioned.

5.6. The energetics of H ⇒ X phase transition

It remains to deal with H ⇒ X transitions, for which no thermochemical data are available. The enthalpy of H ⇒ X transformation in Gd_2O_3 is estimated as a difference between enthalpies of fusion of yttrium and gadolinium sesquioxides under assumption that enthalpies of melting of H-Y_2O_3 and H-Gd_2O_3 (if H ⇒ X transition would not occur) are equal. This can be justified to some extent by the facts that both oxides have similar melting points and heat capacities, while the cations do not show any partial divalent or tetravalent character in the melt [58]. The entropy of H ⇒ X transition in Gd_2O_3 is then 4.569 J mol^{-1} K^{-1} and the same value is assumed for all the rare earth sesquioxides. The enthalpies of H ⇒ X transitions in R_2O_3 (R = La–Dy) are then evaluated using the assessed temperatures of those phase transformations (Table 2.2), whereas reasonable estimates for the enthalpies the hypothetical H ⇒ X transitions in Ho_2O_3, Er_2O_3, Tm_2O_3, Yb_2O_3, Lu_2O_3, and Sc_2O_3 are obtained by extrapolation of the line drawn through the corresponding values for La_2O_3 and Gd_2O_3 versus ionic radius. For Y_2O_3, a slightly higher enthalpy (12,665 J mol^{-1}) than the extrapolated value (12,474 J mol^{-1}) is recommended, as it allows better fit of the phase boundaries in the Y_2O_3–R_2O_3 systems, where R = Nd, Eu, Sm, Gd (see Section 7).

6. Revised thermodynamic functions

The assessed standard enthalpies of formation, standard entropies, enthalpies and entropies of phase transitions as well as the calculated transition temperatures for all rare earth sesquioxides, including metastable modifications are collected in Table 6.1. With an

Table 6.1
Assessed thermodynamic properties of rare earths sesquioxides at ambient pressure ($P = 1$ bar)

Phase	T (K)	$^\circ S$	$\Delta_f{}^\circ H$	Transition	$\Delta_{trs}{}^\circ H$	$\Delta_{trs}{}^\circ S$
C-Sc_2O_3	298.15	77.00	−1,908,573			
	2758			C ⇒ L	100,416	36.409
				C ⇒ B*	31,378	3.058
				C ⇒ A*	50,000	5.720
				C ⇒ H*	28,829	9.654
				C ⇒ X*	42,868	14.223
C-Y_2O_3	298.15	98.96	−1,932,800			
	2600			C ⇒ H	25,100	9.654
	2712			H ⇒ L	83,679	30.855
				C ⇒ B*	11,087	3.967
				C ⇒ A*	18,569	5.439
				C ⇒ X*	37,765	14.223
A-La_2O_3	298.15	127.38	−1,791,600			
	2313			A ⇒ H	32,350	13.986
	2373			H ⇒ X	10,842	4.569
	2578			X ⇒ L	98,137	38.067
				A ⇒ C*	8337	−7.788
				A ⇒ B*	4139	−2.215
A-Ce_2O_3	298.15	148.80	−1,792,380			
	2338			A ⇒ H	32,699	13.986
	2413			H ⇒ X	11,025	4.569
	2503			X ⇒ L	89,963	35.942
				A ⇒ C*	4390	−7.232
				A ⇒ B*	2283	−1.961
A-Pr_2O_3	298.15	152.74	−1,809,900			
	2353			A ⇒ H	32,909	13.986
	2443			H ⇒ X	11,162	4.569
	2573			X ⇒ L	95,423	37.086
				A ⇒ C*	1320	−6.847
				A ⇒ B*	839	−1.798
A-Nd_2O_3	298.15	157.89	−1,806,900			
	2373			A ⇒ H	33,189	13.986
	2473			H ⇒ X	11,300	4.569
	2593			X ⇒ L	99,132	38.23
				A ⇒ C*	−1311	−6.550
				A ⇒ B*	−399	−1.684
C-Pm_2O_3	298.15	155.60	−1,811,000			
	457			C ⇒ B	2151	4.709
	936			B ⇒ A	1499	1.602
	2390			A ⇒ H	33,427	13.986
	2498			H ⇒ X	11,414	4.569
	2593			X ⇒ L	102,098	39.375
C-Sm_2O_3	298.15	145.19	−1,826,798			
B-Sm_2O_3	298.15	151.01	−1,822,998			
	676			C ⇒ B	3312	4.899
	2173			B ⇒ A	3324	1.5295
	2403			A ⇒ H	33,608	13.986

Table 6.1 (*continued*)

Phase	T (K)	$^\circ S$	$\Delta_f{}^\circ H$	Transition	$\Delta_{trs}{}^\circ H$	$\Delta_{trs}{}^\circ S$
	2523			H ⇒ X	11,528	4.569
	2608			X ⇒ L	105,673	40.519
C-Eu_2O_3	298.15	134.61	−1,656,582			
B-Eu_2O_3	298.15	142.74	−1,650,400			
	894.5			C ⇒ B	4395	4.914
	2323			B ⇒ A	3494	1.504
	2413			A ⇒ H	33,748	13.986
	2543			H ⇒ X	11,619	4.569
	2623			X ⇒ L	109,282	41.663
C-Gd_2O_3	298.15	150.61	−1,827,531			
B-Gd_2O_3	298.15	157.37	−1,819,700			
	1425			C ⇒ B	5325	3.737
	2443			B ⇒ A	6300	2.579
	2481			A ⇒ H	34,700	13.986
	2633			H ⇒ X	12,031	4.569
	2693			X ⇒ L	71,702	26.625
C-Tb_2O_3	298.15	156.91	−1,865,200			
	1823			C ⇒ B	7676	4.211
	2440			B ⇒ H	34,126	13.986
	2643			H ⇒ X	12,076	4.569
	2683			X ⇒ L	75,273	28.056
				C ⇒ A*	13,027	5.668
C-Dy_2O_3	298.15	149.79	−1,863,400			
	2263			C ⇒ B	9255	4.090
	2463			B ⇒ H	34,448	13.986
	2653			H ⇒ X	12,122	4.569
	2681			X ⇒ L	79,051	29.486
				C ⇒ A*	15,637	5.546
C-Ho_2O_3	298.15	158.15	−1,883,300			
	2458			C ⇒ B	9787	3.982
	2483			B ⇒ H	34,727	13.986
	2688			H ⇒ L	83,102	30.916
				C ⇒ A*	17,131	5.451
				C ⇒ X*	56,969	22.537
C-Er_2O_3	298.15	155.64	−1,900,100			
	2593			C ⇒ H	25,033	9.654
	2691			H ⇒ L	87,044	32.346
				C ⇒ B*	12,464	3.885
				C ⇒ A*	20,703	5.380
				C ⇒ X*	37,614	14.223
C-Tm_2O_3	298.15	139.72	−1,889,300			
	2623			C ⇒ H	25,322	9.654
	2698			H ⇒ L	84,404	31.284
				C ⇒ B*	13,968	3.800
				C ⇒ A*	23,031	5.328
				C ⇒ X*	38,019	14.223

(*continued on next page*)

Table 6.1 (*continued*)

Phase	T (K)	$°S$	$\Delta_f°H$	Transition	$\Delta_{trs}°H$	$\Delta_{trs}°S$
C-Yb_2O_3	298.15	133.05	−1,814,500			
	2653			C ⇒ H	25,612	9.654
	2708			H ⇒ L	81,839	30.221
				C ⇒ B*	15,345	3.725
				C ⇒ A*	25,165	5.293
				C ⇒ X*	38,415	14.223
C-Lu_2O_3	298.15	109.96	−1,877,000			
	2763			C ⇒ L	80,566	29.159
				C ⇒ B*	16,598	3.659
				C ⇒ A*	27,106	5.270
				C ⇒ H*	27,000	9.654
				C ⇒ X*	39,899	14.223

Enthalpies are in J mol^{-1} and entropies are in J mol^{-1} K^{-1}. Estimated data for metastable modifications are marked with stars.

exception of B-Sm_2O_3, B-Eu_2O_3, and B-Gd_2O_3, the first column in this Table shows the only polymorphic modification, which is thermodynamically stable at $T = 298.15$ K and $P = 1$ bar and the standard entropy and enthalpy of formation refer to this phase. The standard enthalpies of formation of A-La_2O_3, A-Pr_2O_3, A-Nd_2O_3, C-Pm_2O_3, C-Sm_2O_3, B-Sm_2O_3, B-Eu_2O_3, B-Gd_2O_3, C-Tb_2O_3, C-Dy_2O_3, C-Ho_2O_3, C-Er_2O_3, C-Tm_2O_3, C-Yb_2O_3, and C-Lu_2O_3 are listed as recommended by Cordfunke and Konings [6]. Enthalpy of formation of A-Ce_2O_3 has been evaluated in the course of thermodynamic assessment of the Ce–O system [110]. The standard entropies of A-La_2O_3, A-Ce_2O_3, A-Pr_2O_3, and A-Nd_2O_3 are taken from the recent evaluation by Gruber et al. [7], while the values for the other lanthanide sesquioxides (B-Sm_2O_3, B-Eu_2O_3, C-Gd_2O_3, C-Dy_2O_3, C-Ho_2O_3, C-Er_2O_3, C-Tm_2O_3, C-Yb_2O_3, C-Lu_2O_3) are accepted as reported in original publications [70,72,73,76]. For Pm_2O_3 and Tb_2O_3 only estimated entropy data are available [111]. The standard entropy and enthalpy of formation of Y_2O_3 and C-Sc_2O_3 are adopted from the corresponding critical assessment works [59,60]. The enthalpy of formation of C-Eu_2O_3 and C-Gd_2O_3 as well as the entropy of C-Sm_2O_3, C-Eu_2O_3, and B-Gd_2O_3 are estimated in the present work together with enthalpies and entropies of phase transitions as described in the previous section.

Table 6.2 contains the coefficients of the Gibbs energy equation for rare earth sesquioxides, which are evaluated using the data in Table 6.1 in conjunction with the available measurements of heat capacity and/or enthalpy increment. Pm_2O_3 is not included in Table 6.2, since no corresponding data are reported. The Gibbs energy functions for other polymorphic modifications can easily be obtained by adding the enthalpy and entropy of the corresponding phase transition to the coefficients a, and b, respectively. Since the enthalpies cannot be defined absolutely, the Gibbs energy is referred to the constant enthalpy values of the so-called stable element references, H^{SER}, at 298.15 K and 1 bar [112]. For Sm_2O_3, Eu_2O_3 and Gd_2O_3, heat capacities, which can be derived from the Gibbs energy expressions are then directly compared with the experimental data (see below). For the corresponding enthalpy increment data, only differences between the experimental and calculated values are plotted *versus* temperature. In this way, discrepancies in the available data can easily be detected. They may have different origins, most probably different cal-

Table 6.2
The coefficients of the Gibbs energy equation for rare earth sesquioxides, referred to the stable element reference: $°G(\mathrm{J\ mol^{-1}}) = a + bT + cT\ln T + \mathrm{d}T^2 + eT^{-1} + fT^{-2}$

Phase	a	b	c	d	e	f
C-Sc_2O_3	−1,955,472	751.4893	−120.843	−0.004,227	1,826,000	-5.2×10^7
C-Y_2O_3	−1,976,462	731.6512	−121.881	−0.005,060	1,090,000	-1.3×10^7
A-La_2O_3	−1,833,257	692.9664	−120.629	−0.006,854	808,000	-1.0×10^7
A-Ce_2O_3	−1,832,858	667.3306	−119.855	−0.012,155	546,000	0
A-Pr_2O_3	−1,851,548	683.3800	−122.780	−0.012,360	638,000	-1.0×10^7
A-Nd_2O_3	−1,847,329	637.4243	−116.358	−0.014,677	711,000	-1.0×10^7
C-Sm_2O_3	−1,875,835	780.6356	−135.618	−0.006,896	1,191,000	0
B-Sm_2O_3	−1,871,213	751.7711	−132.137	−0.008,367	1,405,000	-4.0×10^7
C-Eu_2O_3	−1,701,762	774.5807	−133.819	−0.009,091	667,000	0
B-Eu_2O_3	−1,693,456	732.3929	−128.827	−0.010,086	559,000	0
C-Gd_2O_3	−1,868,812	660.0623	−119.1688	−0.006,438	772,000	0
B-Gd_2O_3	−1,858,111	620.0992	−114.534	−0.007,203	540,000	0
C-Tb_2O_3	−1,903,004	640.3000	−117.520	−0.012,240	250,000	0
C-Dy_2O_3	−1,902,316	679.1313	−122.593	−0.006,971	59,000	4.0×10^7
C-Ho_2O_3	−1,921,338	654.7487	−120.347	−0.005,998	91,000	3.0×10^7
C-Er_2O_3	−1,939,714	652.6562	−119.216	−0.005,889	428,000	2.0×10^7
C-Tm_2O_3	−1,927,401	680.6164	−121.506	−0.005,899	0	4.0×10^7
C-Yb_2O_3	−1,853,511	702.7502	−123.821	−0.004,567	0	5.0×10^7
C-Lu_2O_3	−1,920,290	710.2224	−120.356	−0.004,579	1,144,000	-2.0×10^7

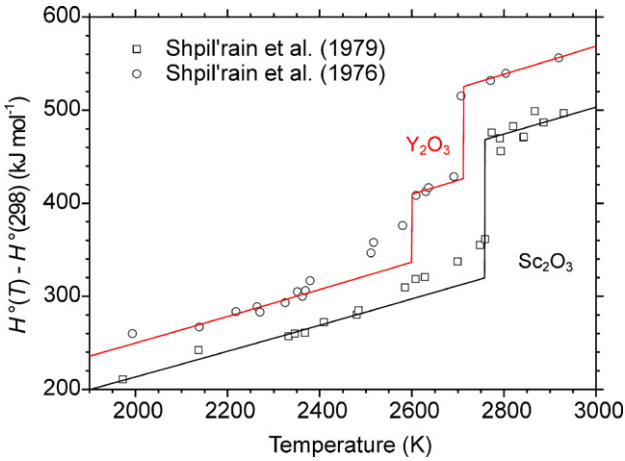

Fig. 6.1. The high-temperature enthalpy increments for Sc_2O_3 and Y_2O_3, experimental data (squares, circles) and values calculated from the Gibbs energy expressions in Table 6.2 (lines, shifted on the vertical scale).

ibrations of calorimeters (in terms of temperature and enthalpy) and different impurity levels.

The thermodynamic functions in Table 6.2 are valid between 250 and 3000 K. No change of the heat capacity at any phase transition is assumed due to the lack of data, except for C ⇒ B transformations in Sm_2O_3, Eu_2O_3, and Gd_2O_3. This also means that the heat capacities of the liquid rare earth sesquioxides are described by the same functions

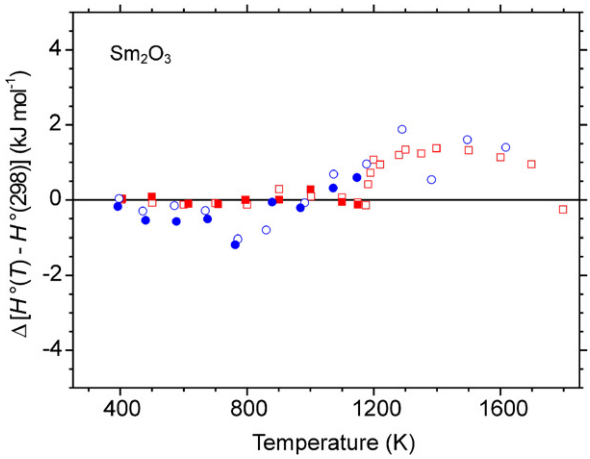

Fig. 6.2. The differences between experimental and calculated enthalpy increment of C-Sm_2O_3 (filled symbols) and of B-Sm_2O_3 (open symbols). Squares and circles correspond to the data from Refs. [80,88], respectively.

as for the high-temperature solid phases, respectively, whereas the available measurements for the liquid Sc_2O_3 and Y_2O_3 [16,66] indicate the heat capacity of about 200 J mol^{-1} K^{-1}, which is either constant or decreasing with temperature. However, due to the scatter and the limited temperature range an uncertainty of the heat capacity values derived from these data may be very large. Fig. 6.1 shows the high-temperature enthalpy increments for yttrium and scandium sesquioxides: both measured [16,66] and calculated from the Gibbs energy expressions (Table 6.2). It can be seen that the experimental data for the liquid phases are also well described by the heat capacity functions for the solid sesquioxides. In theory, the heat capacity of a liquid does attain a constant value at sufficiently high

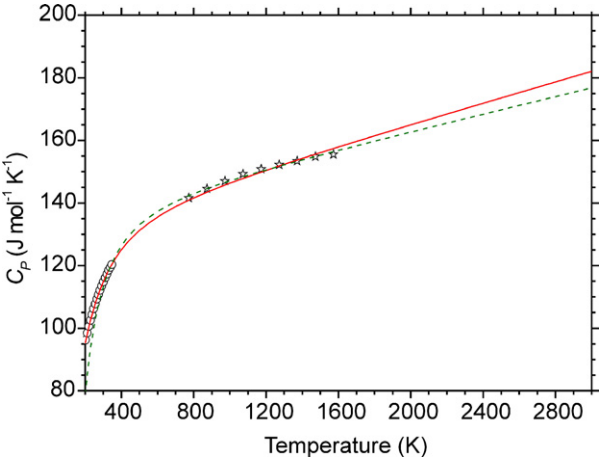

Fig. 6.3. Assessed heat capacity of Sm_2O_3 in comparison with the experimental data for B-modification: Refs. [72] (circles) and [65] (stars). Solid and dashed curves correspond to the present description of B- and C-Sm_2O_3, respectively.

temperatures, but if this constant heat capacity is extrapolated below the melting point a temperature would eventually be reached, where the entropy of the liquid becomes lower than the entropy of the stable crystalline phase (so-called Kauzmann paradox). It is evident that such a situation may never arise if a single function for the heat capacity is used in solid and liquid regions, as recommended in the present work. In addition, it is commonly observed that below melting point, the heat capacity of a supercooled liquid approaches the heat capacity of a solid as the temperature decreases. Since rare earth sesquioxides are refractory compounds, the approximation of the heat capacities of the

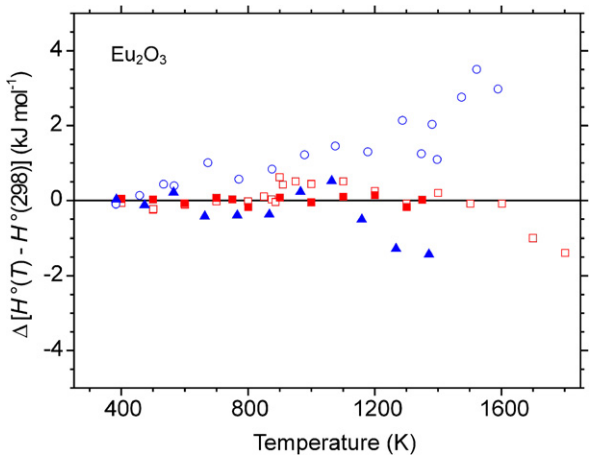

Fig. 6.4. The differences between experimental and calculated enthalpy increment of C-Eu_2O_3 (filled symbols) and of B-Eu_2O_3 (open symbols). Squares, circles, and triangles correspond to the data from Refs. [80,88], and [84], respectively.

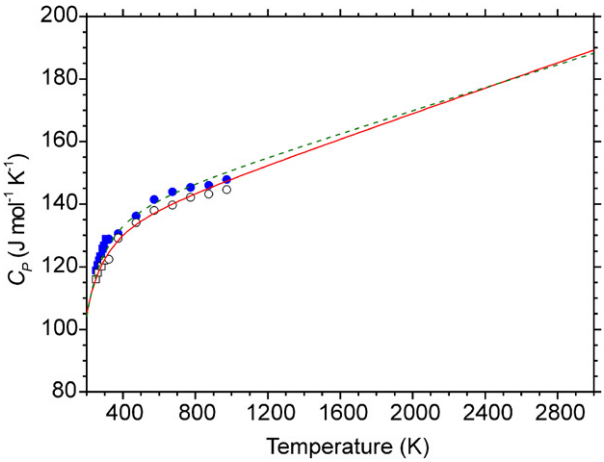

Fig. 6.5. Assessed heat capacity of Eu_2O_3 in comparison with the data from refs. [76] (squares) and [96] (circles). Filled symbols: C-Eu_2O_3, open symbols: B-Eu_2O_3. Solid and dashed curves correspond to the present description of B- and C-Eu_2O_3, respectively.

liquids by the corresponding functions for the high-temperature solid phases is good enough for most applications, such as phase diagram calculations.

6.1. Scandium sesquioxide

Three series of the high-temperature enthalpy measurements are reported in the ranges 394–1799 K [81], 385–1607 K [87], and 1570–2930 K [66]. The data of Shpil'rain et al. [66] show a considerable scatter at lower temperatures, but good agreement with the measurements of Pankratz and Kelley [81] exists between 2332 and 2480 K. These two datasets are

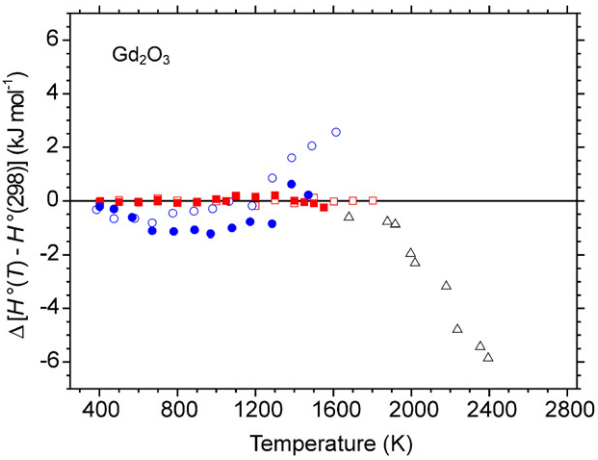

Fig. 6.6. The differences between experimental and calculated enthalpy increment of C-Gd_2O_3 (filled symbols) and of B-Gd_2O_3 (open symbols). Squares, circles, and triangles correspond to the data from Refs. [80,90,97], respectively.

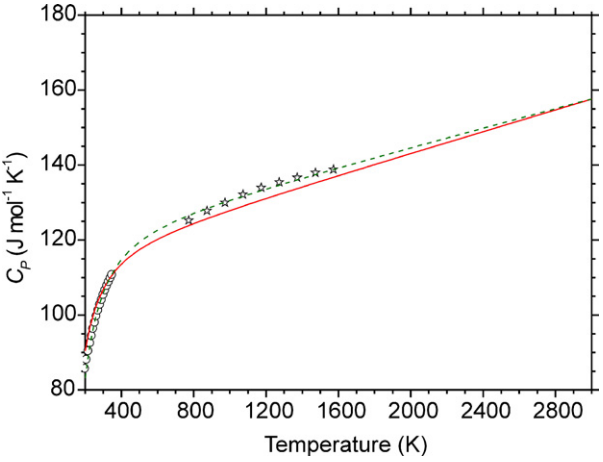

Fig. 6.7. Assessed heat capacity of Gd_2O_3 in comparison with the data from Refs. [72] (circles, C-form) and [65] (stars, B-form). Solid and dashed curves correspond to the present description of B- and C-Gd_2O_3, respectively.

used together with the only available low-temperature heat capacity data [69] to obtain a reasonably smooth fit. Another series of measurements [87] shows increasing positive deviations starting from 780 K and is discarded. In the previous evaluation [59], all three datasets [66,81,87] have been used to derive the heat capacity function, which shows, however, an unreasonable increase at high temperatures.

6.2. Yttrium sesquioxide

Low-temperature heat capacity of C-Y_2O_3 has been measured twice [67,75]. Above room temperature, heat capacity data between 773 and 1573 K are reported [65], whereas

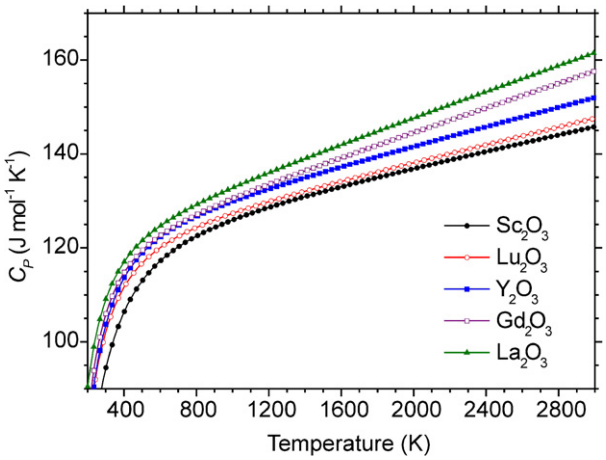

Fig. 6.8. Assessed heat capacities of C-Sc_2O_3, C-Lu_2O_3, C-Y_2O_3, C-Gd_2O_3, and A-La_2O_3 as functions of temperature.

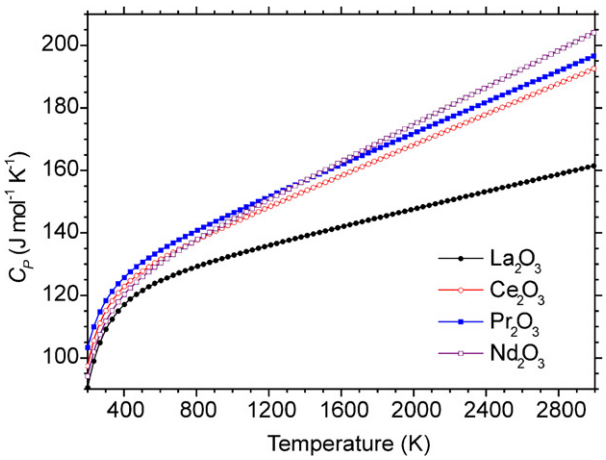

Fig. 6.9. Assessed heat capacities of A-type rare earth sesquioxides: La_2O_3, Ce_2O_3, Pr_2O_3, and Nd_2O_3.

the high-temperature enthalpy measurements have been carried out in the ranges 398–1799 K [80], 389–1615 K [86], 1194–2468 K [98], and 1993–2919 K [16]. The values of the first three works are in fair agreement, whereas the data of Shpil'rain et al. [16] show a systematic positive deviation of about 30 kJ mol^{-1}. However, Pankratz et al. [80] found an unexplained minor thermal anomaly near 1330 K, which may indicate the presence of impurities. The enthalpy increment values reported by Yashvili et al. [86] are preferred as they show the least scatter. The derived heat capacity function perfectly fits the more recent low-temperature heat capacity measurements [75]. Previous assessment [60] relied on the heat capacity data of Kolitsch [65], which are not in good agreement with either

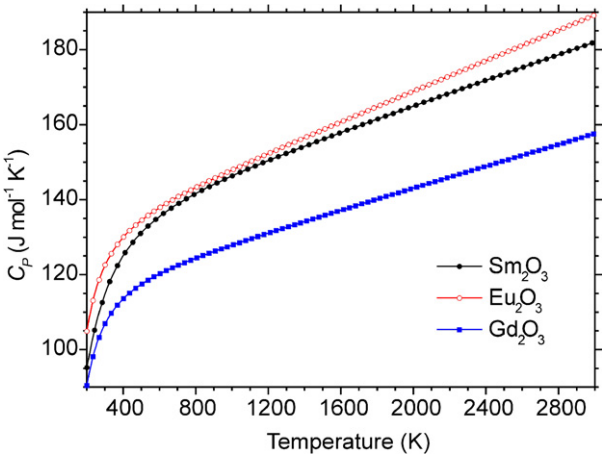

Fig. 6.10. Assessed heat capacities of B-type rare earth sesquioxides: Sm_2O_3, Eu_2O_3, and Gd_2O_3.

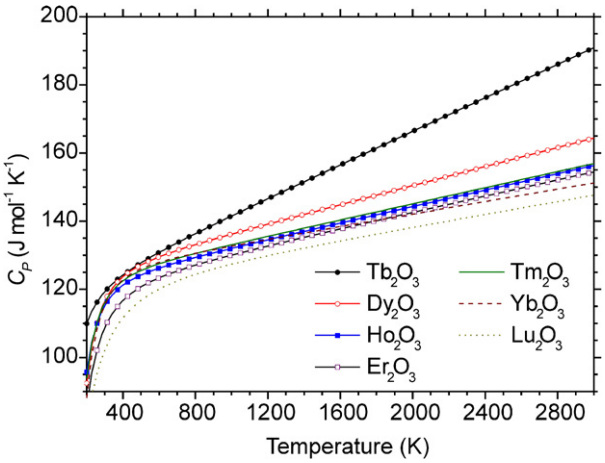

Fig. 6.11. Assessed heat capacities of C-type rare earth sesquioxides: Tb_2O_3, Dy_2O_3, Ho_2O_3, Er_2O_3, Tm_2O_3, Yb_2O_3, and Lu_2O_3.

high-temperature enthalpy values. The function proposed by Landa et al. [94] based on their own measurements between 1300 and 2100 K gives even higher heat capacity for this temperature range.

6.3. Lanthanum sesquioxide

Low-temperature heat capacity measurements for A-La_2O_3 have been made by three groups [67,68,71]. High-temperature enthalpy measurements are available for the temperature ranges 384–1172 K [78], 399–1797 K [68], and 390–1626 K [89]. In addition, heat capacity of A-La_2O_3 has been determined within the temperature limits 400–850 K [95] and 773–1573 K [65]. All enthalpy increments data are in good accord and agree well with the low-temperature data of Justice and Westrum [71], while both high-temperature heat capacity datasets [65,95] show small but systematic deviations from the derived function. The heat capacity curve optimized by Grundy et al. [113] is very close to the present description, but shows increasing positive deviations above 1000 K.

6.4. Cerium sesquioxide

Two series of the high-temperature enthalpy measurements for A-Ce_2O_3 are available, which cover the ranges 400–1000 K [83] and 470–883 K [77]. There are large discrepancies between these datasets, likely due to the different amounts of mixed oxide phase of cerium, which inevitably presents in all Ce_2O_3 samples [7]. A number of heat capacity measurements are reported in the low- [69,74,77] and high-temperature (400–1100 K) [95] regions. According to the present evaluation, the heat capacity of Ce_2O_3 is at best represented by the data of Weller and King [69] and Huntelaar et al. [77]. The heat capacity description proposed by Kuznetsov and co-workers [79] is apparently oversimplified and cannot be used in a wide temperature range.

6.5. Praseodymium sesquioxide

No low-temperature heat capacity data for Pr_2O_3 have been published. However, Gruber et al. [7] quoted the corresponding measurements by Lyutsareva et al. as personal communication. High-temperature enthalpy increments are given by single dataset [85] and refer to A-Pr_2O_3. The derived heat capacity function is essentially based on these data.

6.6. Neodymium sesquioxide

There are two datasets for the low-temperature heat capacity of A-Nd_2O_3 [67,71]. Above room temperature, heat capacity data between 773 and 1573 K are reported [65], whereas the high-temperature enthalpy measurements have been carried out in the ranges 384–1172 K [78] and 400– 1795 K [80]. The values of these two works are in good agreement. However, Pankratz et al. [80] found an unexplained minor thermal anomaly near 1395 K, which may indicate the presence of impurities, so that their data above this temperature are discarded. The derived heat capacity function fits very well the low-temperature measurements of Justice and Westrum [71], whereas the high-temperature heat capacity data of Kolitsch [65] cannot be fully reconciled with the enthalpy increment values, although they also show good compliance between 1273 and 1473 K.

6.7. Samarium sesquioxide

Heat capacity data are available only for B-Sm_2O_3 in the low- [72] and high-temperature (773–1573 K) [65] regions, respectively. High-temperature enthalpy measurements have been carried out for B-Sm_2O_3 in the ranges 404–1798 K [80] and 396–1617 K [88] as well as for C-Sm_2O_3 in the ranges 406–1149 K [80] and 391–1147 K [88]. The two series of enthalpy increment data are in fair agreement (Fig. 6.2) despite the minor thermal anomaly found for B-Sm_2O_3 by Pankratz et al. [80] near 1195 K. However, the data of this work show less scatter and better agree with the heat capacity measurements [65,72]. Consequently, they are used to derive the heat capacity functions for C- and B-Sm_2O_3 (Fig. 6.3) with the cut-off temperature of 1195 K for the latter. It can be seen that heat capacities of two modifications are very similar and nearly identical at 298 K. It is interesting to note, however, that the heat capacity of C-Sm_2O_3 is slightly higher than that of B-Sm_2O_3 below 1236 K, whereas the opposite behaviour is observed above this temperature.

6.8. Europium sesquioxide

Heat capacities of both C- and B-modifications of Eu_2O_3 are available in the low-temperature region [76] as well as between 323 and 973 K [96]. The agreement between these two datasets at the intersection is quite good. High-temperature enthalpy measurements have been carried out for B-Eu_2O_3 in the ranges 400–1802 K [80] and 383–1589 K [88] as well as for C-Eu_2O_3 in the ranges 400–1350 K [80] and 384–1371 K [84]. The data of Pankratz et al. [80] are preferred because of less scatter (Fig. 6.4) despite the minor thermal anomaly found for B-Eu_2O_3 near 895 K. The derived heat capacity functions (Fig. 6.5) fit very well the available data, that would be impossible, if the data of Gvelesiani et. al. [88] were used. At room temperature, the heat capacity of C-Eu_2O_3 is slightly higher than that of B-Eu_2O_3, while the crossover occurs around 2477 K.

6.9. Gadolinium sesquioxide

Low-temperature heat capacity measurements have been made for C-Gd_2O_3 [72], while high-temperature (773–1573 K) heat capacity data are available for B-Gd_2O_3 [65]. High-temperature enthalpy measurements have been carried out for B-Gd_2O_3 in the ranges 400–1802 K [80], 385–1613 K [90], and 1680–2395 K [97] as well as for C-Gd_2O_3 in the ranges 403–1550 K [80] and 401–1472 K [90]. Among the enthalpy increments data, the values measured by Tsagareishvili et al. [90] show very strange wavy behaviour, whereas those of Shpil'rain et al. [97] show systematic deviations, which become more pronounced with increasing temperature (Fig. 6.6). The measurements of Pankratz et al. [80] show very little scatter and they are used to derive the heat capacity functions for both C- and B-modifications. A good fit of the cryogenic heat capacities [72] is obtained (Fig. 6.7), while the high-temperature data of Kolitsch [65] are well consistent with the present description of C-Gd_2O_3. The behaviour of the heat capacity functions with temperature continues the trend found in Sm_2O_3 and Eu_2O_3 (see above), i.e., heat capacity of the cubic polymorph is initially higher than that of the monoclinic phase. The variation of the crossover temperature is also very reasonable, being 1236, 2477, and 3019 K for Sm_2O_3, Eu_2O_3, and Gd_2O_3, respectively.

6.10. Terbium sesquioxide

No heat capacity measurements are reported. High-temperature enthalpy increments are given by single dataset [85] and refer to C-Tb_2O_3. The derived heat capacity function is essentially based on these data.

6.11. Dysprosium sesquioxide

Heat capacity data for C-Dy_2O_3 are only available in the low-temperature region [70]. High-temperature enthalpy measurements have been carried out in the ranges 417–1801 K [83] and 389–1621 K [93]. The latter dataset is preferred, since it better fits the low-temperature values and because of unexplained minor thermal anomaly near 1590 K, found by Pankratz et al. [83].

6.12. Holmium sesquioxide

Heat capacity data for C-Ho_2O_3 are available in the low- [70] and high-temperature (773–1573 K) [65] regions, respectively. High-temperature enthalpy measurements have been carried out in the ranges 400–1799 K [82] and 397–1621 K [91]. The latter dataset is again preferred, since it better fits the low-temperature values. The high-temperature heat capacity data of Kolitsch [65] are also consistent with the derived function.

6.13. Erbium sesquioxide

Heat capacity data for C-Er_2O_3 are only available in the low-temperature region [70]. High-temperature enthalpy measurements have been carried out in the ranges 399–1797 K [82] and 387–1625 K [92]. The former dataset is preferred, because of the better compliance with the low-temperature values.

6.14. Thulium sesquioxide

Heat capacity data for C-Tm_2O_3 are only available in the low-temperature region [73]. High-temperature enthalpy measurements have been carried out in the ranges 397–1801 K [82] and 397–1607 K [84]. Both datasets are in very good agreement below 1000 K, but show increasing divergence at higher temperatures. The derived heat capacity function is therefore based on the low-temperature parts of enthalpy increment values. Unfortunately, no very good fit of the low-temperature heat capacity can be obtained, although the deviations are not very large.

6.15. Ytterbium sesquioxide

Heat capacity data for C-Yb_2O_3 are available in the low- [72] and high-temperature (773–1573 K) [65] regions, respectively. High-temperature enthalpy measurements have been carried out in the ranges 400–1798 K [82] and 384–1587 K [84]. Both datasets are in fair agreement despite the minor thermal anomaly near 1365 K, found by Pankratz et al. [82]. The derived heat capacity function accounts well for the enthalpy increment data, except for the last few points, but again does not perfectly reproduce the

low-temperature values. The high-temperature heat capacity data of Kolitsch [65], however, deviate substantially from the present description.

6.16. Lutetium sesquioxide

Heat capacity data for Lu_2O_3 are only available in low-temperature region [73]. High-temperature enthalpy measurements have been carried out in the ranges 401–1802 K [83] and 394–1602 K [89]. The former dataset is preferred, because of the better compliance with the low-temperature values.

6.17. Comparative analysis of the heat capacity functions

The derived heat capacity functions for all rare earth sesquioxides are grouped together in Figs. 6.8–6.11 according to the electronic configuration of the cation and crystal structure. Apart from the vibrational component (lattice heat capacity), an important contribution to the heat capacity of many rare earth sesquioxides results from promotion of electrons to higher levels (Schottky effect). However, the heat capacity of Sc_2O_3 ($3p^6$), Y_2O_3 ($4p^6$), La_2O_3($4d^{10}$), and Lu_2O_3 ($4f^{14}$) should be entirely determined by the lattice vibrations. Also, the "electronic" heat capacity of cubic Gd_2O_3 ($4f^7$) has been shown to be negligible above 100 K [114]. The corresponding functions are compared in Fig. 6.8. It can be seen that the heat capacity increases in the order $Sc_2O_3 < Lu_2O_3 < Y_2O_3 < Gd_2O_3 < La_2O_3$, i.e., with increasing ionic radius. This conforms to the theoretical expectations, since the amplitude of oscillations increases with the size of atom. It is worth noting that the curves corresponding to C-type sesquioxides of scandium, lutetium, and yttrium have very similar shape, thus confirming that the heat capacity of these compounds is determined primarily by atomic number of the rare earth element.

Fig. 6.9 shows the heat capacity functions for A-type sesquioxides. According to Gruber et al. [7], the lattice heat capacity decreases in the order $La_2O_3 > Ce_2O_3 > Pr_2O_3 > Nd_2O_3$ that is reasonable in view of the lanthanide contraction. Obviously, the Schottky contribution to the heat capacity of Pr_2O_3 is larger than in the case of Ce_2O_3 (Fig. 6.9) in agreement with the theoretical calculations [7]. Since the "electronic" heat capacity of Nd_2O_3 [114] at 300 K is comparable with that of Ce_2O_3, one might expect the total heat capacity of neodymium sesquioxide falling between the values given by the corresponding functions of lanthanum and cerium sesquioxides. However, at higher temperatures this is not the case and above 1400 K, the heat capacity of Nd_2O_3 exceeds even that of Pr_2O_3 (Fig. 6.9) indicating a possible substantial increase of the Schottky contribution.

Heat capacity functions of B-type rare earth sesquioxides are compared in Fig. 6.10. The heat capacity increases a little from Sm_2O_3 to Eu_2O_3 and then decreases again for Gd_2O_3. This is consistent with the substantial Schottky contribution to the heat capacity of Sm_2O_3 [114], which exceeds that of Pr_2O_3. The corresponding effect in Eu_2O_3 may be even larger, but no theoretical analysis has been made so far. Fig. 6.11 shows the heat capacity functions for C-type sesquioxides, from Tb_2O_3 to Lu_2O_3. Terbium sesquioxide possesses the highest heat capacity among the others. It can be seen that there is a continuous decrease of the heat capacity in the order $Tb_2O_3 > Dy_2O_3 > Ho_2O_3 > Er_2O_3$ (the last two compounds show very similar values) that can be understood as a combined effect of decreasing both the ionic radius and the Schottky contribution [73]. However, the "elec-

tronic" heat capacity increases again from Er_2O_3 to Tm_2O_3 and then slightly decreases for Yb_2O_3 [73,114] that is reflected in Fig. 6.11. In summary, a comparison of the heat capacity functions for different rare earth sesquioxides gives a quite reasonable picture, which is also consistent with the electronic energy levels of lanthanide ions, thus providing additional confidence of the proper selection of data, from which the functions are derived (see above).

7. Calculated $R'_2O_3 - R''_2O_3$ phase diagrams

In order to validate the assessed thermodynamic parameters, especially for the metastable phases (Tables 6.1 and 6.2), a number of R'_2O_3–R''_2O_3 phase diagrams (R′, R″ represent different rare earth cations) are calculated in the present work. For this purpose, only those systems are selected, where sufficient experimental information is available and no double oxides (e.g., $LaYbO_3$, Dy_3ScO_6) are formed. It is emphasized that the goal here is not to reproduce the details of the phase diagrams, but to evaluate the reliability of the estimates. Thus, a simple regular solution model is applied to each phase (φ) in the form:

$$G^{\varphi} = x_{R'_2O_3}\,{}^{o}G^{\varphi}_{R'_2O_3} + x_{R''_2O_3}\,{}^{o}G^{\varphi}_{R''_2O_3} + RT(x_{R'_2O_3}\ln x_{R'_2O_3} + x_{R''_2O_3}\ln x_{R''_2O_3}) + x_{R'_2O_3}x_{R''_2O_3}\,{}^{0}L^{\varphi}_{R'_2O_3,R''_2O_3} \tag{7.1}$$

where ${}^{o}G^{\varphi}_{R'_2O_3}$, ${}^{o}G^{\varphi}_{R''_2O_3}$ are the Gibbs energies of the pure rare earth sesquioxides in stable or metastable form, $x_{R'_2O_3}$, $x_{R''_2O_3}$ stand for the mole fractions of components, R is the gas constant, T is the absolute temperature (K), and ${}^{0}L^{\varphi}_{R'_2O_3,R''_2O_3}$ is so-called *interaction parameter*, which accounts for the deviation from the ideal mixing behaviour. In the liquid state all the rare earth sesquioxides are assumed to mix ideally. The optimum values of interaction parameters determined by the least-squares method [115] are compiled in Table 7.1.

Table 7.1
Optimum values of interaction parameters (J mol^{-1}) according to the regular solution model for selected R'_2O_3–R''_2O_3 systems determined by the least-squares method.

R′	R″	Phase					
		A	B	C	H	X	Liquid
Y	Er	0	0	0	0	0	0
Y	Dy	0	0	0	0	0	0
Sc	Yb	–	–	+25,300	+19,000	–	0
Sc	Er	–	–	+41,400	+28,100	–	0
Y	Gd	−3100	+700	−1950	+1100	+650	0
Y	Eu	−2300	+2400	− 2100	+4800	+3300	0
Y	Sm	+3500	+7950	+1050	+6100	+4500	0
Y	Nd	+16,600	+10,800	+7400	+13,160	+9590	0
Yb	Nd	+16,600	+7400	+11,900	+16,900	+12,600	0
La	Nd	+12,900	–	–	+9000	+7650	0
La	Sm	+17,750	+10,000	+12,000	+11,200	+8900	0
La	Gd	+26,000	+18,200	+20,000	+15,850	+12,700	0
La	Dy	+34,900	+35,900	+40,000	+27,900	+23,400	0
La	Ho	+31,900	+27,700	+30,000	+22,450	+16,750	0

Figs. 7.1 and 7.2 show phase diagrams of the Y_2O_3–Er_2O_3 and Y_2O_3–Dy_2O_3 systems, respectively. No interaction parameters are introduced in both cases, assuming ideal mixing between Y^{+3} and Er^{+3} or Dy^{+3} cations. Indeed, their radii are very close. It can be seen that there is a perfect match between experimental and calculated phase boundaries, except for the region close to Y_2O_3, due to the systematic deviation between the measured temperature of C $\Longleftrightarrow$ H transition in Y_2O_3 and the value accepted in the present work (see also Section 2.1.5). Note that the corresponding two-phase fields are very narrow, and could not be resolved experimentally. In the Y_2O_3–Dy_2O_3 system (Fig. 7.2), there

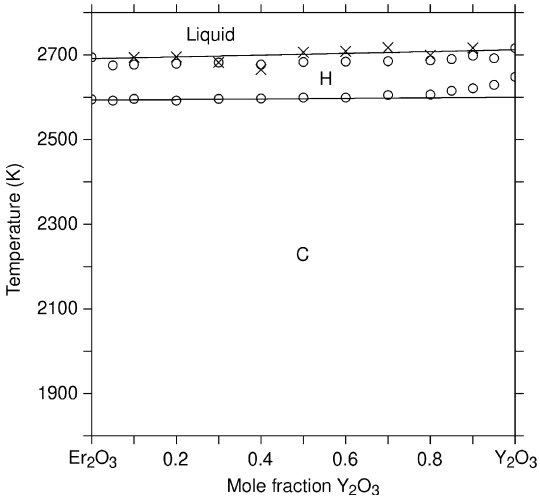

Fig. 7.1. Calculated Er_2O_3–Y_2O_3 phase diagram in comparison with the experimental thermal analysis data [34].

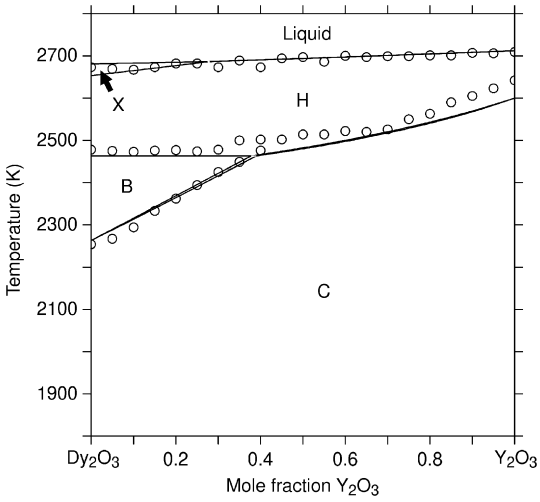

Fig. 7.2. Calculated Dy_2O_3–Y_2O_3 phase diagram in comparison with the experimental thermal analysis data [33].

are two invariant reactions of undetermined type, which correspond to B + C + H and X + H + liquid three-phase equilibria.

Phase diagrams of the Sc_2O_3–Yb_2O_3 and Sc_2O_3–Er_2O_3 systems are shown in Figs. 7.3 and 7.4, respectively. Although there is a complete miscibility in the solid and liquid states, the ideal solution approximation holds only for the liquid phase, whereas a substantial repulsive interactions exist in the C- and H-phases, due to the large differences in the sizes of Sc^{+3} and Yb^{+3} or Er^{+3} ions. The lattice mismatch is larger in the case of erbium substitution, that is reflected in the larger positive interaction parameters (Table 7.1). The

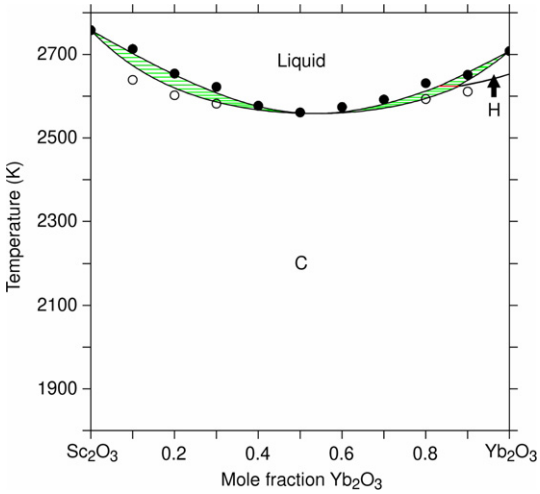

Fig. 7.3. Calculated Sc_2O_3–Yb_2O_3 phase diagram superimposed with the experimental thermal analysis data [102].

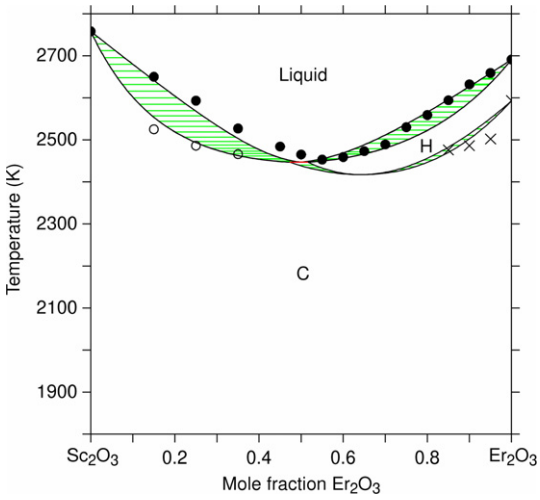

Fig. 7.4. Calculated Sc_2O_3–Er_2O_3 phase diagram superimposed with the experimental thermal analysis data [102].

agreement with experiment is very good taking the uncertainty of the temperature measurements (±40 K) into account. The congruent melting point of C-type solid solution in the Sc_2O_3–Yb_2O_3 system merges with C + H + liquid three-phase equilibrium in the Sc_2O_3–Er_2O_3 system. For the latter, the calculations confirm the prediction of the Badie et al. [102] about the existence of invariant minimum point, which corresponds to the congruent C $\Longleftrightarrow$ H transformation (Fig. 7.4).

Phase diagrams of the systems Y_2O_3–Sm_2O_3, Y_2O_3–Eu_2O_3, and Y_2O_3–Gd_2O_3 are presented in Figs. 7.5–7.7. It can be seen that liquid and H-phase form continuous solutions,

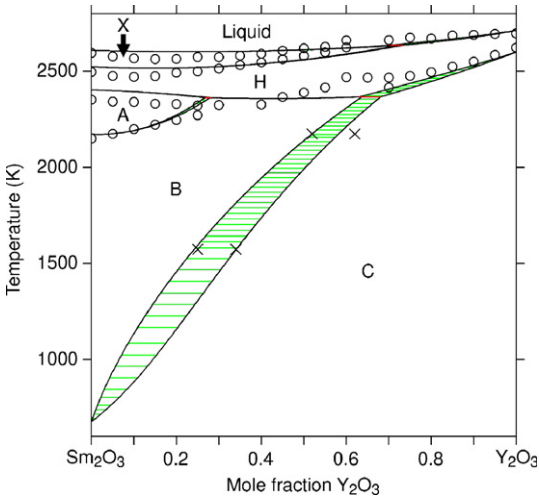

Fig. 7.5. Calculated Sm_2O_3–Y_2O_3 phase diagram with corresponding experimental points [32]. Circles: DTA study, crosses: method of annealing and quenching.

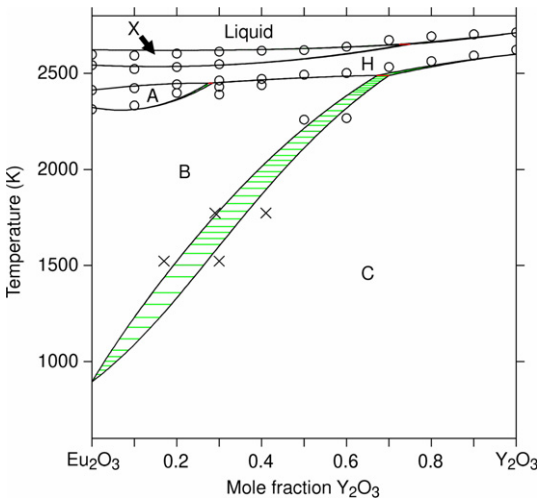

Fig. 7.6. Calculated Eu_2O_3–Y_2O_3 phase diagram with corresponding experimental points [19]. Circles: DTA study, crosses: method of annealing and quenching.

while the solubility of yttrium in A- and X-phases as well as the width of the B + C two-phase field change in the systematic way. This information is used to determine the equilibrium temperatures of C $\Longleftrightarrow$ B phase transitions in Sm_2O_3, Eu_2O_3, Gd_2O_3 (see Section 2.1.1) and enthalpy of H $\Rightarrow$ X transformation in Y_2O_3 (see Section 5.6), since all three systems, Y_2O_3–Sm_2O_3, Y_2O_3–Eu_2O_3, and Y_2O_3–Gd_2O_3 show weak interaction energies, in the order of few kilojoules per mole (Table 7.1). The interaction parameters of B-, H-, and X-phases remain positive and decrease in the order Sm > Eu > Gd, as the difference

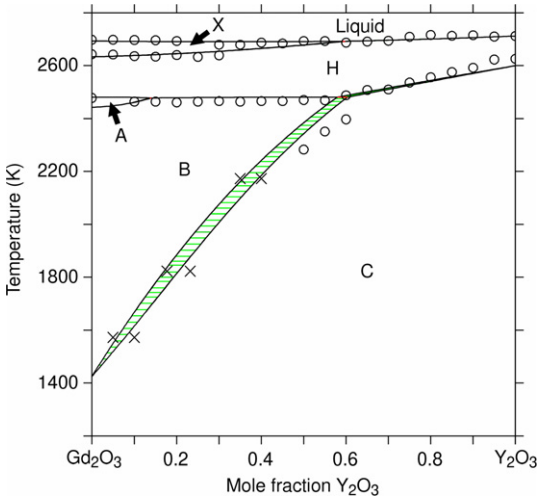

Fig. 7.7. Calculated Gd_2O_3–Y_2O_3 phase diagram with corresponding experimental points [32]. Circles: DTA study, crosses: method of annealing and quenching.

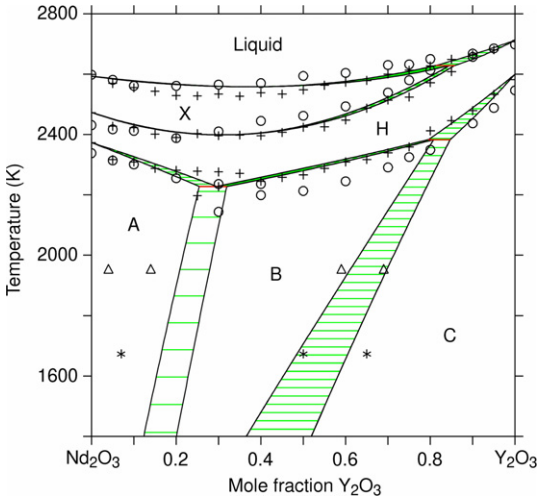

Fig. 7.8. Calculated Nd_2O_3–Y_2O_3 phase diagram in comparison with experimental measurements. Crosses [55] and circles [17]: thermal analysis data, stars [55] and triangles [17]: method of annealing and quenching.

between ionic radii of those elements and Y becomes smaller. For the A- and C-phases the interaction parameters change sign from positive (Sm) to the negative (Eu, Gd), indicating some tendency for ordering. The calculated phase diagrams show good compliance with the available experimental data, neglecting the small differences between assessed and measured temperatures of phase transformations in pure sesquioxides. In each system, three invariant reactions take place, which correspond to A + B + H, B + C + H, and H + X + liquid three-phase equilibria.

Figs. 7.8 and 7.9 show phase diagrams of the Nd_2O_3–Y_2O_3 and Nd_2O_3–Yb_2O_3 systems, respectively. Again, liquid and H-phase form continuous solutions, while yttrium shows

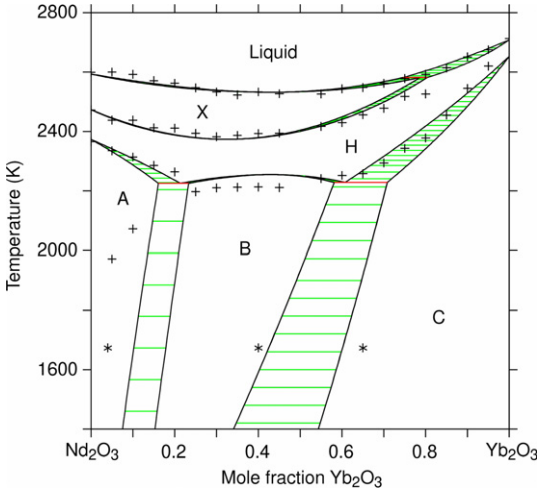

Fig. 7.9. Calculated Nd_2O_3–Yb_2O_3 phase diagram in comparison with experimental data [55]. Crosses: thermal analysis data, stars: method of annealing and quenching.

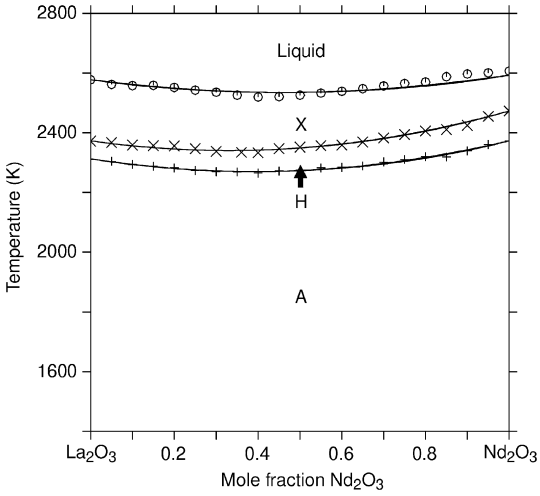

Fig. 7.10. Calculated La_2O_3–Nd_2O_3 phase diagram superimposed with experimental thermal analysis data [12].

higher solubility in the X-phase than ytterbium. In both cases, the X-type solid solution is formed via peritectic reaction. The remarkable feature of these phase diagrams is the appearance of B-type solid solution, which is metastable for every pure sesquioxide although the enthalpy of mixing is positive for all solid phases (Table 7.1). Accordingly, there are two eutectoid- or peritectoid-type invariant reactions in the Nd_2O_3-rich and Y_2O_3/Yb_2O_3-rich sides of the phase diagrams, which correspond to A + B + H and B + C + H three-phase equilibria, whereas the X $\Longleftrightarrow$ liquid and H $\Longleftrightarrow$ X transformations show congruent character. Owing to the smaller ionic radius of Yb^{+3} cation, the homogeneity ranges of C-type solid solution enlarges at the expense of A- and B-type phases. The width of B + C two-phase field is also larger in the case of Nd_2O_3–Yb_2O_3 system, in accordance with the more positive value for the enthalpy of C $\Rightarrow$ B phase transition (Table 6.1). At high temperatures, the agreement between calculations and experimental data is very good, while visible discrepancies exist with respect to the homogeneity ranges of A-, B-, and C-type solid solutions. It should be noted that simple regular solution model may not be applicable even to the R'_2O_3–R''_2O_3 systems, when the differences in ionic radii of R′ and R″ become large enough. The observed crossover from negative to positive enthalpy of mixing for the C-phase in the Nd_2O_3–Y_2O_3 system [103] supports this conclusion.

Phase diagram of the La_2O_3–Nd_2O_3 system is presented in Fig. 7.10. All phases form continuous solutions. Perfect agreement between calculations and experimental data is evident. Figs. 7.11 and 7.12 show phase diagrams of the La_2O_3–Sm_2O_3 and La_2O_3–Gd_2O_3 systems, respectively. These diagrams are topologically similar: there is a complete miscibility between La_2O_3 and Sm_2O_3 or Gd_2O_3 in the A-, H-, X-, and liquid phases, whereas the A + B two-phase field is larger in the case of La_2O_3– Gd_2O_3 system, as a consequence of increasing lattice mismatch and the enthalpy of B $\Rightarrow$ A phase transition (Table 6.1). The available experimental data are well reproduced by calculations neglecting the small differences between assessed and measured temperatures of B $\Rightarrow$ A and A $\Rightarrow$ H transformations in gadolinium sesquioxide. The increasing dissimilarity between La^{3+} and other cations

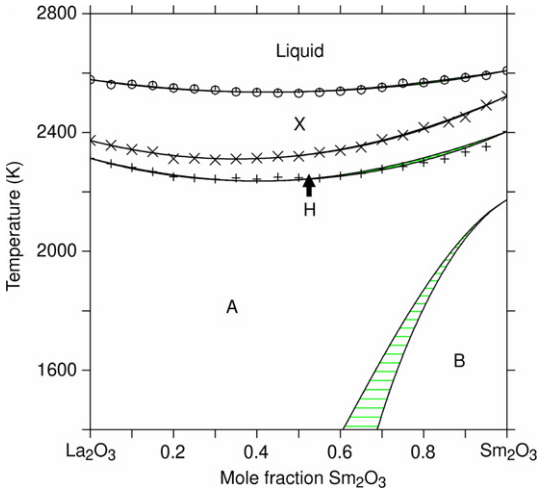

Fig. 7.11. Calculated La_2O_3–Sm_2O_3 phase diagram superimposed with experimental thermal analysis data [12].

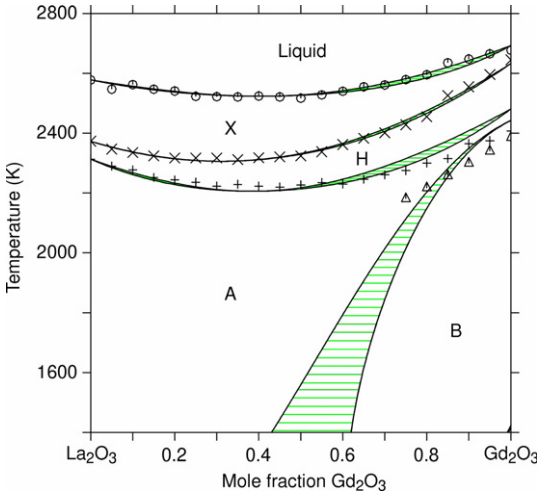

Fig. 7.12. Calculated La_2O_3–Gd_2O_3 phase diagram superimposed with experimental thermal analysis data [12].

(i.e., Nd^{+3}, Sm^{+3}, and Gd^{+3}) is reflected in a systematic increase of the positive interaction parameters in the order Nd < Sm < Gd (Table 7.1) as well as in the deeper minimum points of the congruent X $\Longleftrightarrow$ liquid, H $\Longleftrightarrow$ X, and A $\Longleftrightarrow$ H transformations (Figs. 7.10–7.12). In addition, the A + H, H + X, and X + liquid two-phase fields, which are almost invisible on the La_2O_3–Nd_2O_3 phase diagram (Fig. 7.10) and can hardly be recognized for La_2O_3–Sm_2O_3 system (Fig. 7.11) are clearly seen in the case of La_2O_3–Gd_2O_3 system (Fig. 7.12).

Phase diagrams of the La_2O_3–Dy_2O_3 and La_2O_3–Ho_2O_3 systems are shown in Figs. 7.13 and 7.14, respectively. It can be seen that the trends observed in the systems La_2O_3–Nd_2O_3, La_2O_3–Sm_2O_3, and La_2O_3–Gd_2O_3 are also continued here. However, the

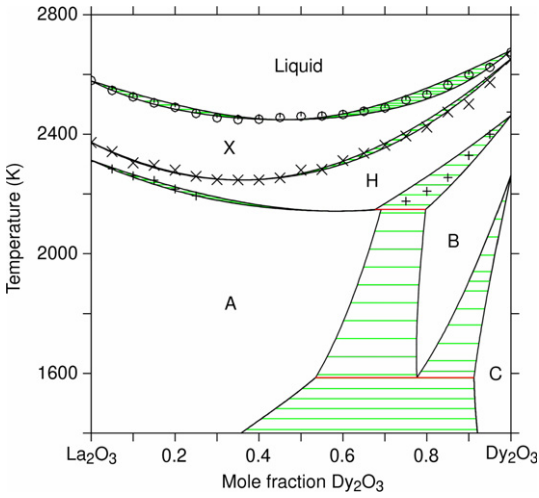

Fig. 7.13. Calculated La_2O_3–Dy_2O_3 phase diagram superimposed with experimental thermal analysis data [12].

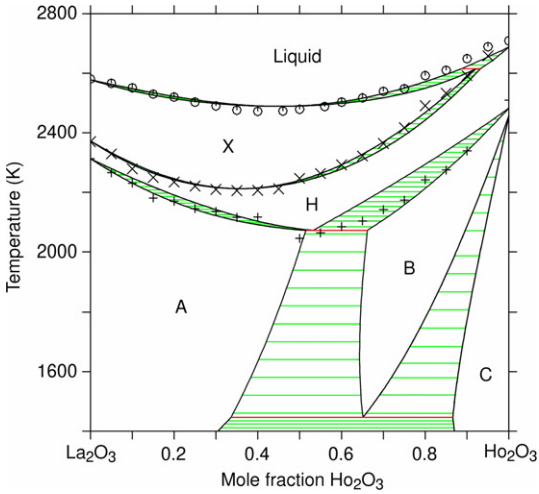

Fig. 7.14. Calculated La_2O_3–Ho_2O_3 phase diagram superimposed with experimental thermal analysis data [12].

new two-phase fields, A + C, B + C, and H + B appear, whereas the minimum points of the congruent A $\Longleftrightarrow$ H transformation are transformed into peritectoid and eutectoid reactions on the La_2O_3–Dy_2O_3 and La_2O_3–Ho_2O_3 phase diagrams, respectively. Owing to the higher instability of A-Ho_2O_3 with respect to B- and C-modifications compared to Dy_2O_3, the homogeneity range of the A-type solid solution shrinks. Additionally, X-phase is metastable for pure holmium sesquioxide, and it decomposes peritectically around 90 mol% Ho_2O_3 (Fig. 7.14). It should be noted that an eutectoidal decomposition of the B-phase, which is seen in Figs. 7.13 and 7.14 also takes place in the La_2O_3–Sm_2O_3 and La_2O_3–Gd_2O_3 systems according to calculations, where the corresponding temperatures are off the plot. The agreement between calculations and experiment is quite satisfactory.

8. Concluding remarks

Reexamination of all known thermodynamic data for rare earth sesquioxides lead to improved revised values of the thermodynamic functions. In addition, using ionic radius of a trivalent rare earth cation as an universal parameter, it was possible to generate a new knowledge concerning the enthalpies and entropies of the polymorphic phase transformations. Reasonable estimates for the relative stabilities of different structures, including metastable modifications as well as for the enthalpies and entropies of melting were thus obtained. It is hoped that all these data will be useful for many practical applications of rare earth sesquioxides, but also for thermodynamic calculations of phase diagrams. Although the rationality of the derived functions and transition data is already demonstrated through the sample calculations of R'_2O_3–R''_2O_3 phase diagrams, the agreement with experimental data can be further improved with more sophisticated models than the regular solution approximation.

Further refinement (verification) of the values suggested in this review is of course possible. This can be done either through the new experimental studies or via first-principles calculations. In addition, the quantitative estimation of the elastic energy contribution

would make it possible to unify various evidences, such as the linear dependencies of the molar volume on ionic radius, the appearance of B-type structure in the intermediate composition range between A- and C- type structures, the dependences of heat capacity on ionic radius, etc. The recommendations for future experimental works are summarized below:

(1) Redetermination of the temperatures of A $\Longleftrightarrow$ H and H $\Longleftrightarrow$ X phase transitions in Ce_2O_3 and Pr_2O_3 under controlled atmosphere to ensure the stoichiometric compositions.
(2) Measurements of the enthalpies of the following polymorphic phase transformations: C $\Rightarrow$ B in Pm_2O_3, Eu_2O_3, Tb_2O_3, Dy_2O_3, and Ho_2O_3, B $\Rightarrow$ A in Pm_2O_3, Sm_2O_3, and Eu_2O_3, A $\Rightarrow$ H in La_2O_3, Ce_2O_3, Pr_2O_3, Nd_2O_3, Pm_2O_3, Sm_2O_3, and Eu_2O_3, B $\Rightarrow$ H in Tb_2O_3, Dy_2O_3, and Ho_2O_3, C $\Rightarrow$ H in Er_2O_3, Tm_2O_3, and Yb_2O_3 as well as H $\Rightarrow$ X in La_2O_3, Ce_2O_3, Pr_2O_3, Nd_2O_3, Pm_2O_3, Sm_2O_3, Eu_2O_3, Gd_2O_3, Tb_2O_3, and Dy_2O_3.
(3) Measurements of the enthalpies of fusion for all lanthanide sesquioxides.
(4) Measurement of the enthalpy of formation of any modification of Pm_2O_3.
(5) Low-temperature heat capacity measurements for Pm_2O_3, C-Sm_2O_3, and Tb_2O_3.
(6) High-temperature enthalpy or heat capacity measurements for all rare earth sesquioxides up to the melting points.

There remain, also, two interesting questions which are not yet answered and may motivate for the further research on rare earth sesquioxides:

(1) Do the B $\Rightarrow$ A phase transitions in R_2O_3 (R = Tb–Lu, Y, Sc) occur reversibly when increasing pressure? Was the transformation into A-phase overlooked in the studies of Hoekstra [36] and Seck et al. [37] or it requires higher pressures?
(2) Does the cubic fluorite type structure represent the high-temperature modification of Y_2O_3? Can the oxygen activity in the gaseous environment induce structural changes in yttria and other rare earth sesquioxides at high temperatures?

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